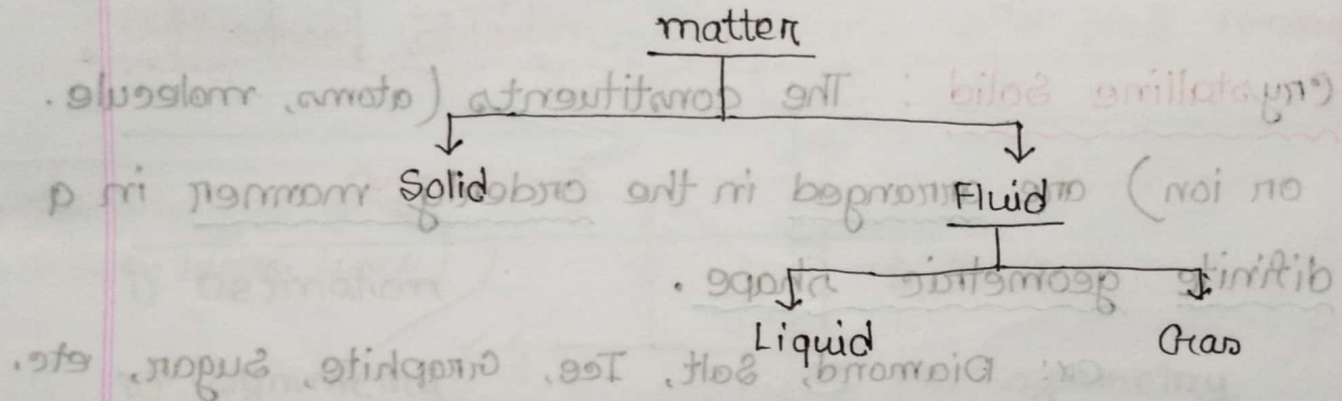


Structure of Matter

— Mamun Sir

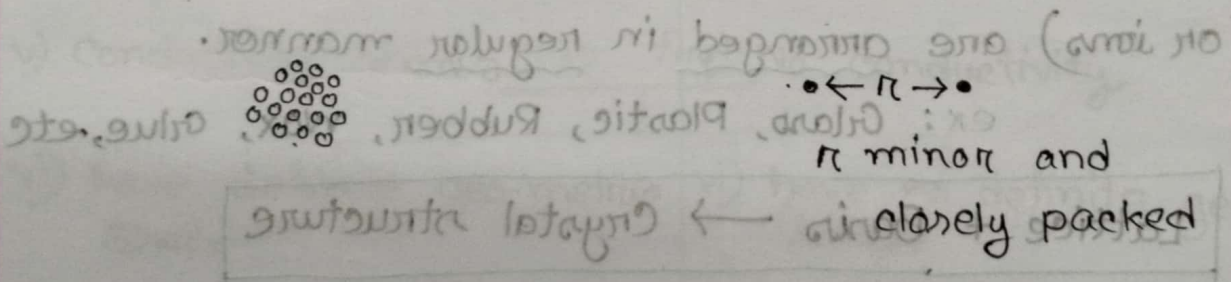
States of matter :

Matter: Matter is anything that has specific mass and occupies specific space.

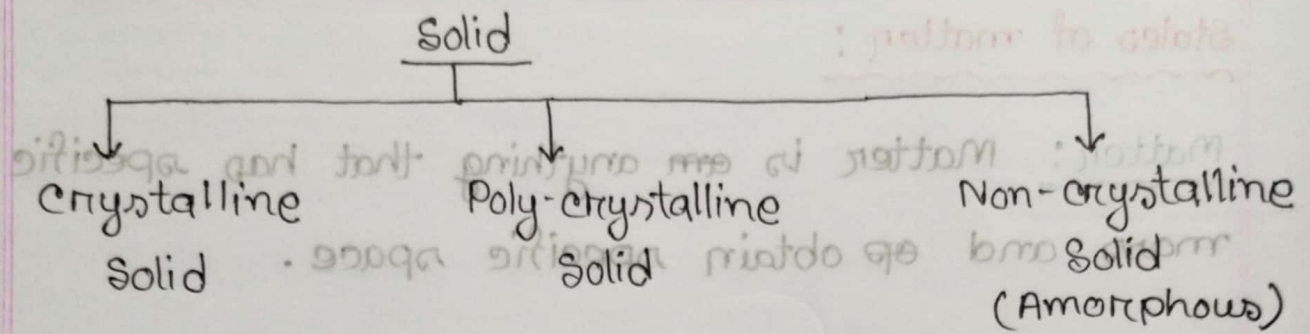


Property of General Solid:

- i) Incompressible
- ii) Inflexible
- iii) have mechanical strength
- iv) have specific space.



A regular and periodic arrangement of particles in space.



Crystalline Solid: The constituents (atoms, molecule or ion) are arranged in the orderly manner in a definite geometric shape.

ex: Diamond, Salt, Ice, Graphite, Sugar, etc.

poly-crystalline solid: Occur as powder resemble amorphous but properties are just like crystalline solid.

ex: Sand, ceramic

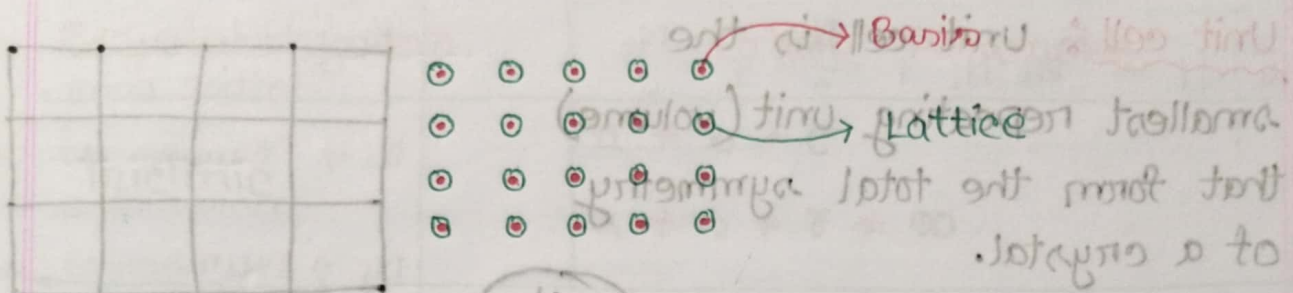
Non-crystalline solid: The constituents (atoms, molecule, or ions) are arranged in regular manner.

ex: Glass, Plastic, Rubber, Wax, Glue, etc

Lattice + Basis $\rightarrow$ Crystal structure

Basis: Basis mean atom or group of atoms.

Lattice: A regular and periodic arrangement of point in space.



Differences between crystalline solid and Amorphous:

Crystalline

Amorphous

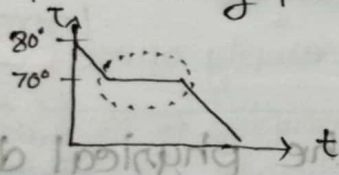
i) Defination

i) Defination

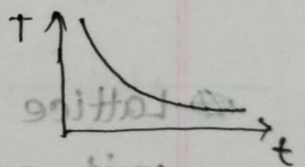
ii) symmetry

ii) No symmetry

iii) Sharp melting point



iii) Do not have sharp melting point



iv) Anisotropic (Direction Dependent)

iv) Isotropic (Direction Independent)

v) Conductivity shows

v) No conductivity

vi) have definite geo-metric shape

vi) have no definite geo-metric shape

vii) Long range order

vii) short range order

viii) heavy Bond energy

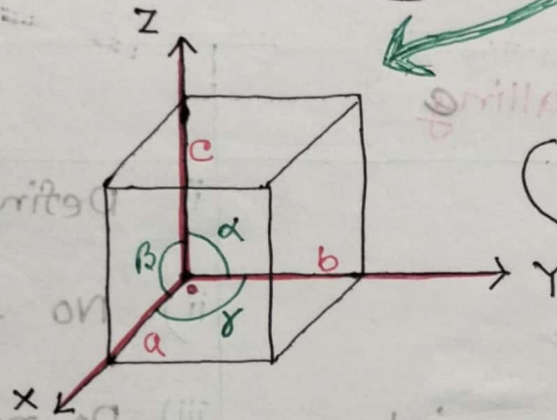
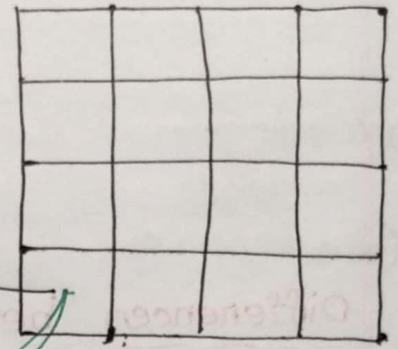
viii) Low bond energy

ix) Example

ix) Example ,

Unit cell : Unit cell is the smallest repeating unit (volume) that form the total symmetry of a crystal.

Unit cell



axial length $\rightarrow a, b, c$
axial angle $\rightarrow \alpha, \beta, \gamma$
Lattice Parameter.

Lattice parameter refers to the physical dimension of unit cell.

Unit cell

Primitive (Simple)

(lattice point will stay at corner)

Non-primitive

(lattice point will stay at corner along with the body)

Crystal system

Lattice Parameter

Triclinic

$$a \neq b \neq c$$

$$\alpha \neq \beta \neq \gamma \neq 90$$

Monoclinic

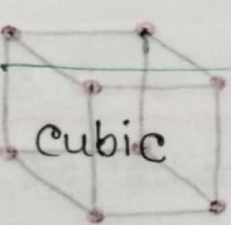
$$a \neq b \neq c$$

$$\alpha = \beta = \gamma \neq 90$$

Orthorhombic

$$a \neq b \neq c$$

$$\alpha = \beta = \gamma = 90$$



Cubic

$$a = b = c$$

$$\alpha = \beta = \gamma = 90$$

Trigonal

$$a = b = c$$

$$\alpha = \beta = \gamma \neq 90$$

Tetragonal

$$a = b \neq c$$

$$\alpha = \beta = \gamma = 90$$

Hexagonal

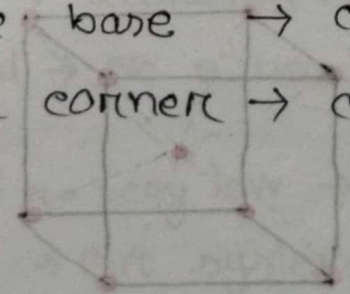
$$a = b \neq c$$

$$\alpha = \beta = 90, \gamma = 120$$

Basis inside of Body $\rightarrow$ contribution 100%

Basis on the base $\rightarrow$ contribution 50%

Basis at the corner $\rightarrow$ contribution $1/8 = 12.5\%$



$$N = N_i + \frac{N_f}{2} + \frac{N_c}{8}$$

$N \rightarrow$ Number of effective lattice point.

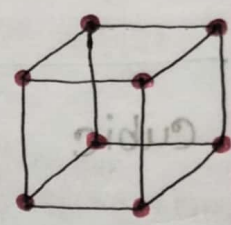
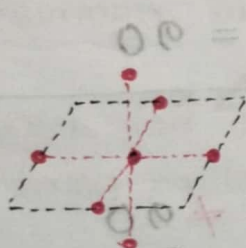
$N_i \rightarrow$ Number of atom inside of a crystal

$N_f \rightarrow$ Number of atom in base / surface

$N_c \rightarrow$ Number of atoms in corner.

Simple cubic : (Primitive)

The crystal structure is cubic and has atom in every corner. Each atom has 6 nearest neighbours.



• lattice point = $\frac{8}{8} = 1$

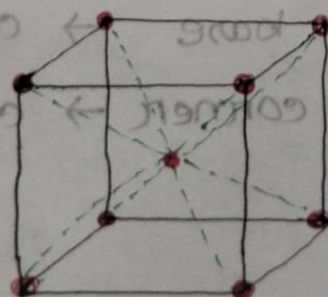
• has more empty space

• has less occupied space

• The co-ordinate number = 6

Non-Primitive

1. Body-centred : The nearest atom to the corner atom is the body-centred atom. A corner atom is surrounded by 8 unit cell and eight centered atom.



• more condensation

• less empty space

• closely packed.

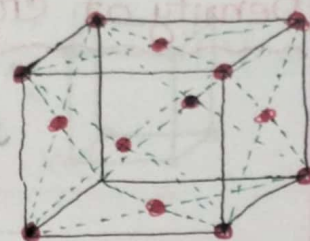
• co-ordinate number = 8

• lattice point = $\frac{8}{8} + 1 = 2$

2. Face-centred :

The lattice point will take place all the 6 faces with 8 corner face.

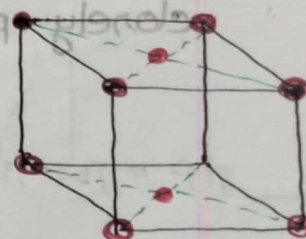
The corner atom has $\frac{1}{8}$ on the same plane, 4 above and 4 below.



- more closely packed
- very less empty space
- co-ordinate number = 12
- Lattice point = $\frac{8}{8} + \frac{6}{2} = 4$

3. Base-centred :

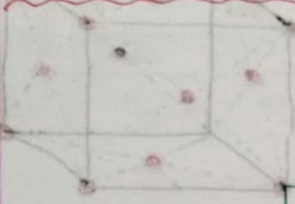
The lattice point will take place at the corner and at the center of top and bottom face only.



- Simple cubic $\rightarrow$ more occupied space $\rightarrow$ closely packed.
- There must be a minimum distance to for closely packed. otherwise system will collapse.
- $\bullet \leftarrow r \rightarrow \bullet \rightarrow r = \infty \rightarrow$ attraction is negligible.
- $\bullet \leftarrow r \rightarrow \bullet \rightarrow r = \text{optimal} \rightarrow$ No e^- repulsion $\rightarrow$ good attraction level
- $\bullet \bullet \rightarrow (r \rightarrow 0) \rightarrow$ if $r = \text{very low} \rightarrow$ No e^- repulsion $\rightarrow$ But system collapse.

Pre-year

Density of crystal lattice:

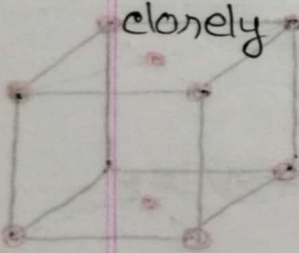


$$\rho = \frac{\text{mass per unit cell}}{\text{volume of unit cell}}$$

$$\rho = \frac{nM}{N_A \cdot abc}$$

$$\begin{aligned} * \text{ mass per unit cell} &= \text{No. of atoms} \times \text{mass of one atom} \\ &= n \times \frac{M}{N_A} \end{aligned}$$

Packing factor: Packing factor indicates how atoms are closely packed in a unit cell.



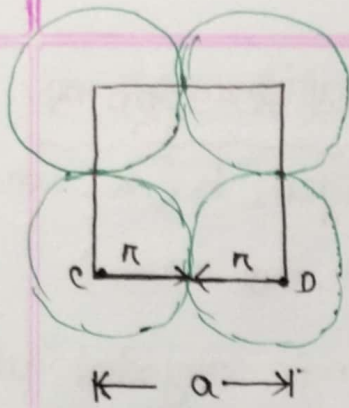
$$P.F = \frac{\text{volume of all atoms in a unit cell}}{\text{volume of unit cell}}$$

$$= \frac{n \times \text{volume of 1 atom}}{abc}$$

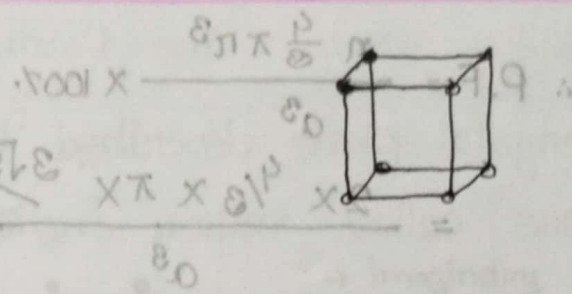
$$= \frac{n \cdot \frac{4}{3} \pi r^3}{abc}$$

$$P.F = \frac{4}{3} \frac{n \pi r^3}{abc}$$

for simple cubic



(Side view)



Now, $CD = a = r + r = 2r$

$$\therefore r = \frac{a}{2}$$

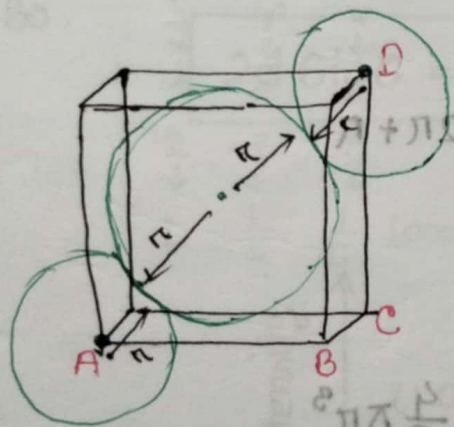
$$\therefore \text{P.F.} = \frac{n \times \frac{4}{3} \pi r^3}{a^3} \times 100\%$$

$$= \frac{1 \times \frac{4}{3} \times \frac{a^3}{8} \times \pi}{a^3} \times 100\%$$

$$= 52\%$$

$\therefore$ Empty space = $(100 - 52) = 48\%$

For Body Centred:-



$$AD = 2r + r + r = 4r \quad \text{--- (1)}$$

Now, $AC^2 = AB^2 + BC^2 = a^2 + a^2$

$$= 2a^2 \quad \text{--- (2)}$$

Now, $AD^2 = AC^2 + CD^2$

$$= 2a^2 + a^2 \therefore AD = \sqrt{3}a$$

$$= 3a^2$$

$$\sqrt{3}a = 4r$$

$$\therefore \sqrt{3}a = 4r$$

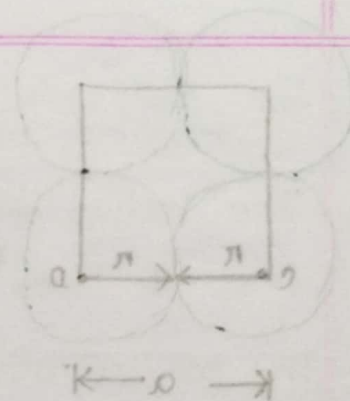
$$\therefore r = \frac{\sqrt{3}}{4}a$$

$$\therefore r = \frac{\sqrt{3}}{4}a$$

$$\therefore P.F. = \frac{n \frac{4}{3} \pi r^3}{a^3} \times 100\%$$

$$= \frac{2 \times \frac{4}{3} \times \pi \times \frac{3\sqrt{3}}{64} \cdot a^3}{a^3} \times 100\%$$

$$= 68\%$$

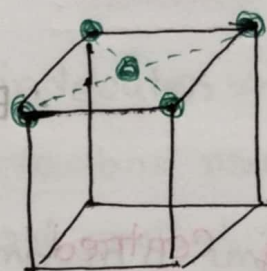
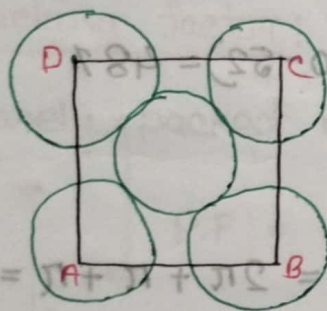


$$\therefore \text{Empty space} = (100 - 68)\% = 32\%$$

$$\frac{\pi \times \frac{4}{3} \times \frac{a}{2} \times \frac{a}{2} \times \frac{a}{2}}{a^3} =$$

$$\frac{r}{a} = \pi \therefore$$

Face centred:



$$AC^2 = AB^2 + BC^2 \quad \therefore AC = \pi + 2\pi + \pi$$

$$= 4\pi$$

$$\Rightarrow AC^2 = 2a^2$$

$$\therefore AC = \sqrt{2} a$$

Now,

$$4\pi = \sqrt{2} a$$

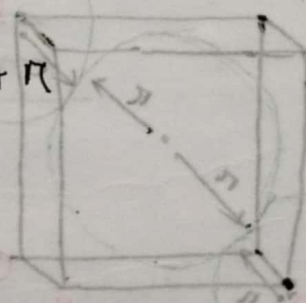
$$\Rightarrow \pi = \frac{\sqrt{2}}{4} a$$

So,

$$P.F. = \frac{n \frac{4}{3} \pi r^3}{a^3} \times 100\%$$

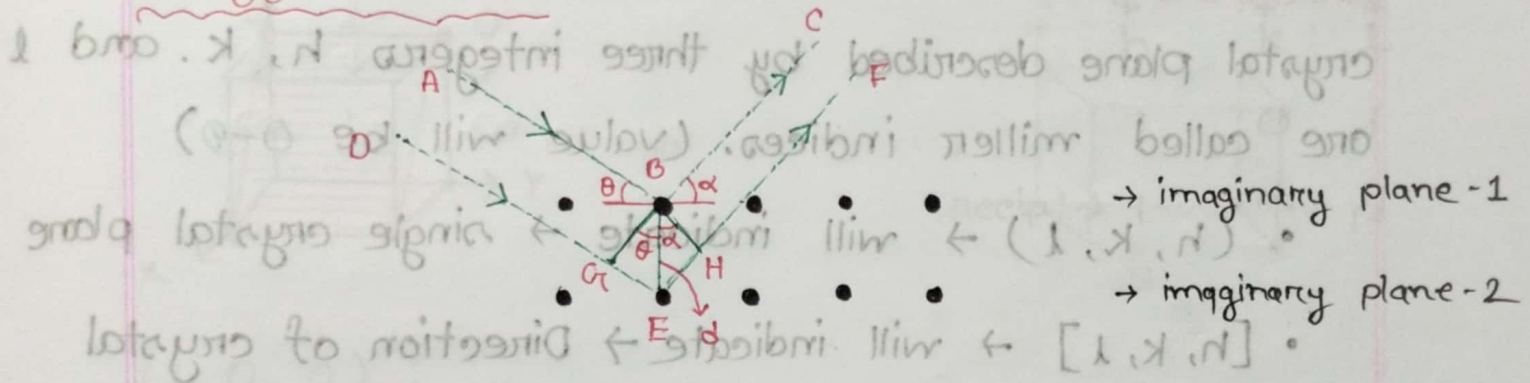
$$= \frac{4 \times \frac{4}{3} \times \pi \times \frac{2\sqrt{2}}{64} a^3}{a^3} \times 100\%$$

$$= 74\%$$



$$\therefore \text{Empty space} = (100 - 74)\% = 26\%$$

Bragger's Law:



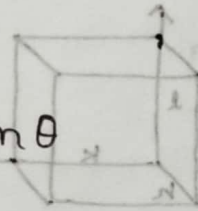
∴ path difference = $GE + EH$

* reciprocal of the integer

= $2GE$

= $2d \sin \theta$

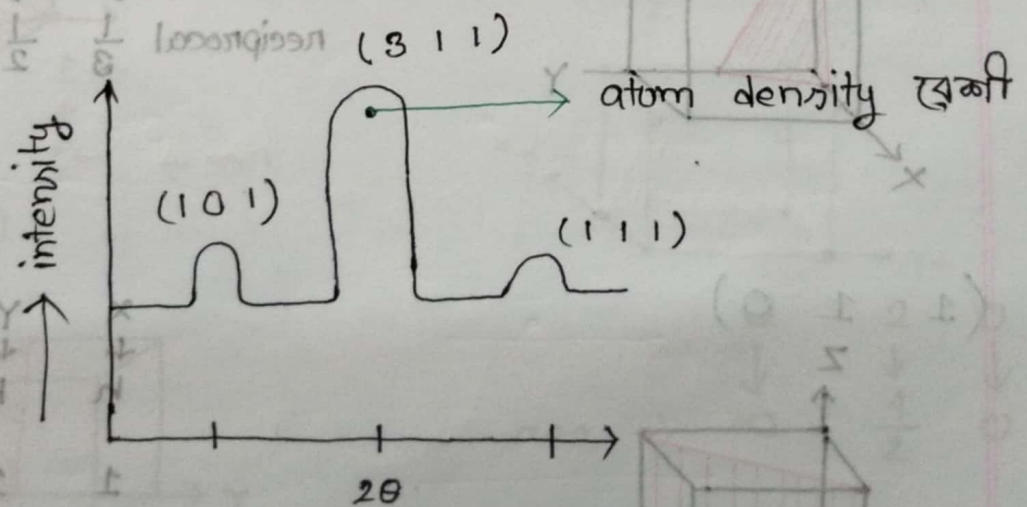
∴ $GE = d \sin \theta$



* To achieve highest intensity peak, path difference has to be $n\lambda$.

So,

$2d \sin \theta = n\lambda$ → Bragg's law



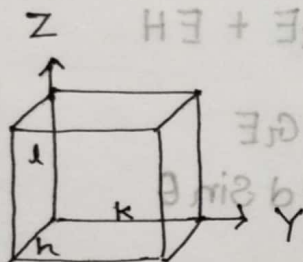
indicates polycrystal

Miller indices: The Position and orientation of a crystal plane described by three integers h , k , and l are called miller indices. (value will be 0-9)

• $(h, k, l) \rightarrow$ will indicate $\rightarrow$ single crystal plane

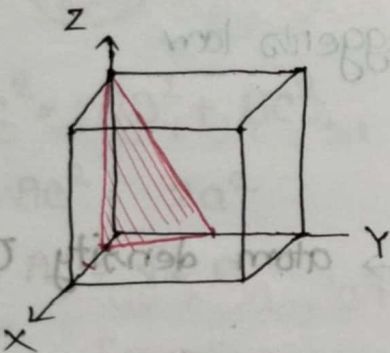
• $[h, k, l] \rightarrow$ will indicate $\rightarrow$ Direction of crystal plane.

* reciprocal will be inverse of the integer.



Plane drawing:

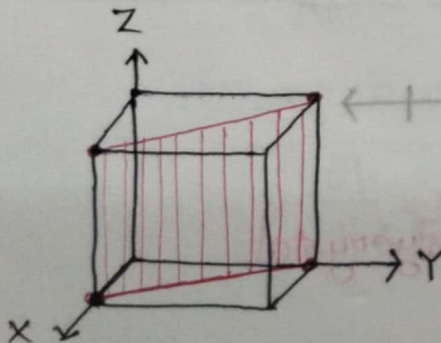
$(3 \ 2 \ 1)$



(1 1 3) reciprocal

X	Y	Z
$\downarrow$ h	$\downarrow$ k	$\downarrow$ l
3	2	1
$\downarrow$	$\downarrow$	$\downarrow$
$\frac{1}{3}$	$\frac{1}{2}$	1

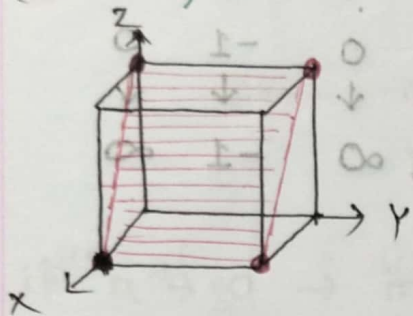
$(1 \ 1 \ 0)$



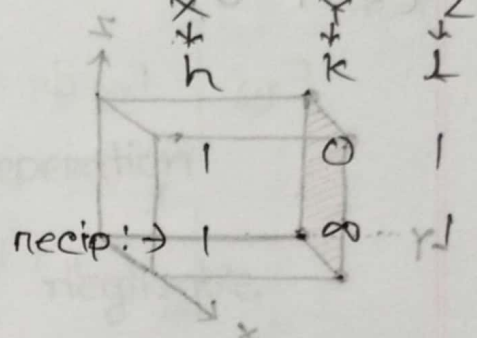
recip $\rightarrow$

X	Y	Z
$\downarrow$ h	$\downarrow$ k	$\downarrow$ l
1	1	0
$\downarrow$	$\downarrow$	$\downarrow$
1	1	∞

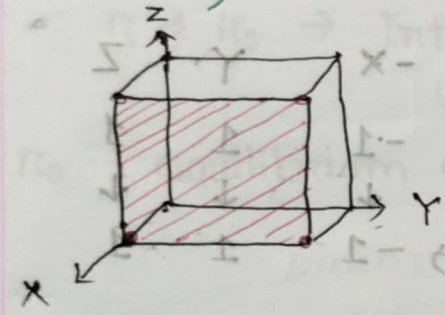
$(1 \ 0 \ 1)$:- X



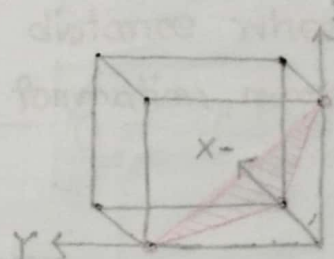
$(0 \ 1 \ 0)$:- Z



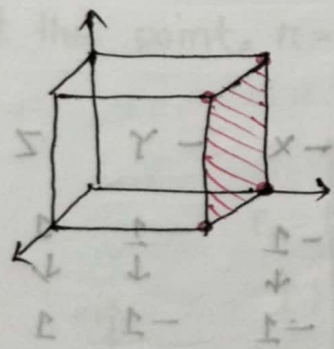
$(1 \ 0 \ 0)$:-



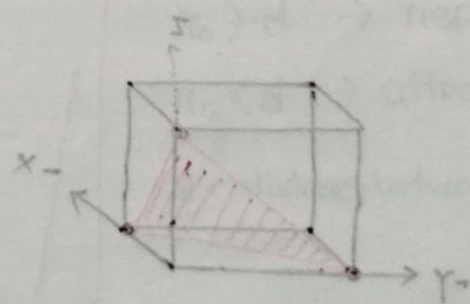
recip $\rightarrow$ $\begin{matrix} 1 & 0 & 0 \\ \downarrow & \downarrow & \downarrow \\ 1 & \infty & \infty \end{matrix}$



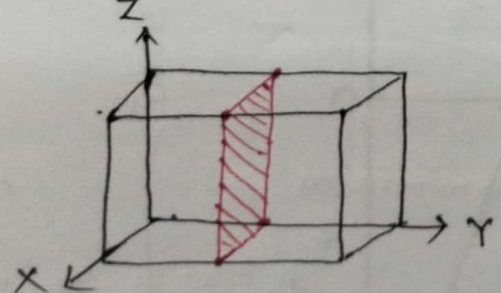
$(0 \ 1 \ 0)$



reciprocal $\rightarrow$ $\begin{matrix} 0 & 1 & 0 \\ \downarrow & \downarrow & \downarrow \\ \infty & 1 & \infty \end{matrix}$

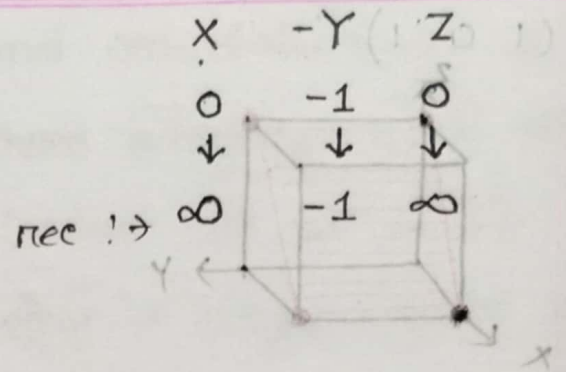
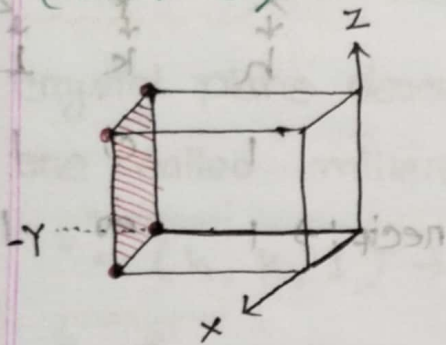


$(0 \ 2 \ 0)$

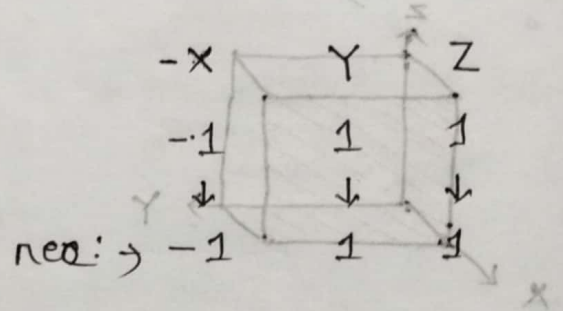
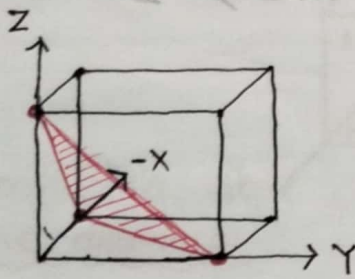


rec $\rightarrow$ $\begin{matrix} 0 & 2 & 0 \\ \downarrow & \downarrow & \downarrow \\ \infty & \frac{1}{2} & \infty \end{matrix}$

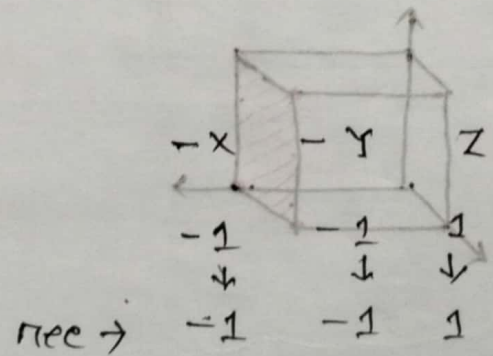
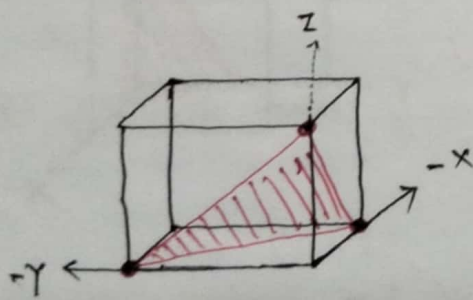
$$* (0 \ 1 \ 0)$$



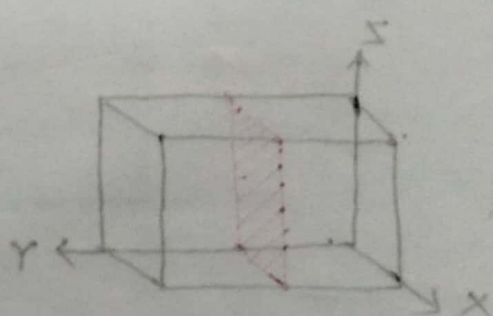
$$\begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} \leftarrow \text{pers}$$

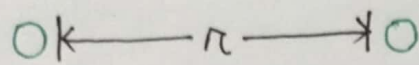


$$\begin{pmatrix} 0 & 1 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \leftarrow \text{pers}$$



$$\begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \leftarrow \text{pers}$$





↳ Inter atomic separation

if $r \rightarrow \infty \rightarrow$ interaction almost negligible

$r \rightarrow 0 \rightarrow$ Interaction highest but system collapse.

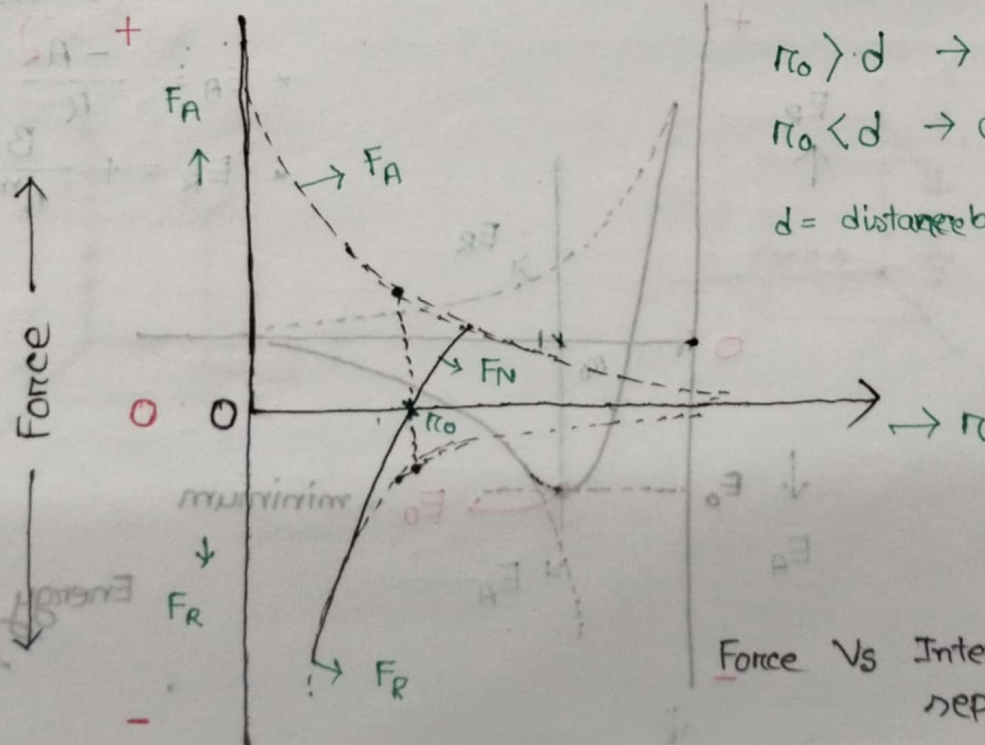
$r \rightarrow r_0 \rightarrow$ Interaction high and system stable.

$r_0 \rightarrow$ equilibrium distance $\rightarrow$ The distance where the bond formation remain stable.

*

$$F_{\text{net}} = F_{\text{Attraction}} + F_{\text{repulsion}}$$

at the point, $r = r_0 \rightarrow F_A = F_R \rightarrow F_{\text{net}} = 0$



$r_0 > d \rightarrow$ repulsive

$r_0 < d \rightarrow$ attractive

$d =$ distance between two atoms

Force Vs Interatomic separation.

Now,

$$E_{\pi} = \int_{\infty}^{\pi} F_{\text{net}} d\pi$$

$$= \int_{\infty}^{\pi} (F_A + F_R) d\pi = \int_{\infty}^{\pi} F_A d\pi + \int_{\infty}^{\pi} F_R d\pi$$

$$= E_A + E_R$$

and,

$$F_N = \frac{dE_{\pi}}{d\pi}, \quad [\pi = \pi_0 \rightarrow F_N = 0]$$

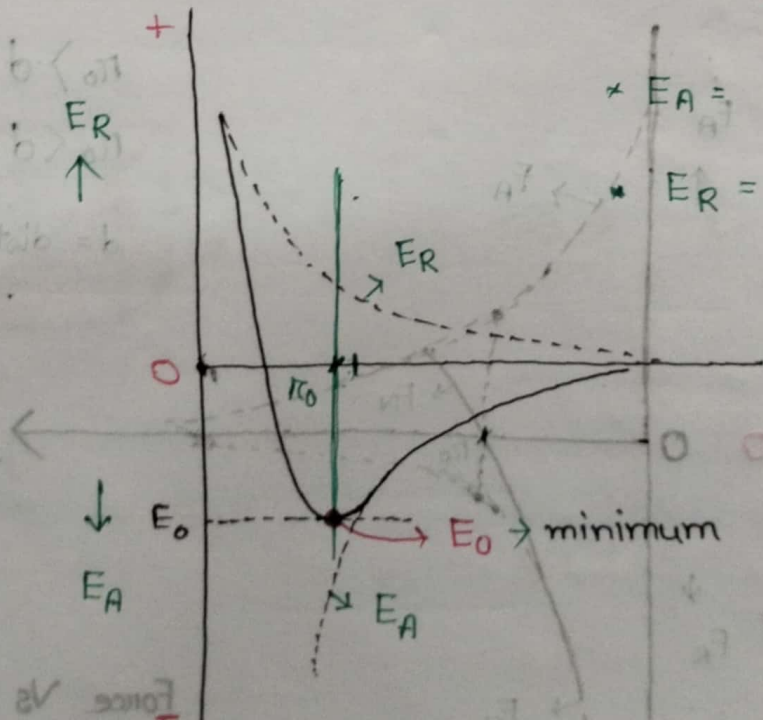
$$\boxed{\frac{dE_{\pi}}{d\pi} = 0} \rightarrow E_{\pi} \text{ passes minima}$$

$$0 = F_{\text{net}} \rightarrow F_A = F_R \rightarrow \pi = \pi_0 \text{ at } F_{\text{net}} = 0$$

$$* E_A = \frac{e^2}{4\pi\epsilon_0 \pi}$$

$$* E_A = \frac{-A}{\pi}$$

$$* E_R = + \frac{B}{\pi^m}$$



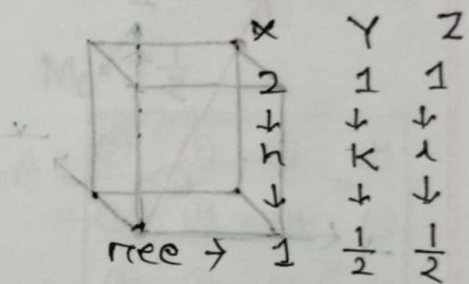
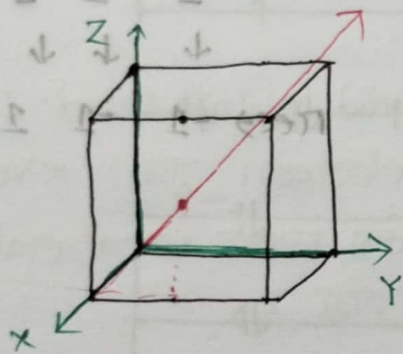
Energy vs Interatomic separation.

Direction of plane:

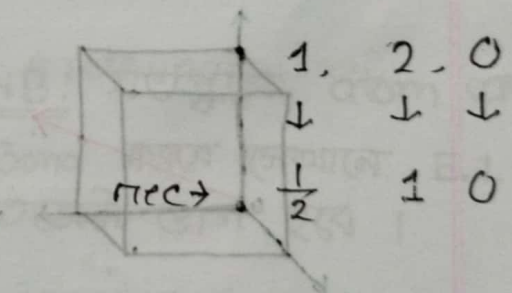
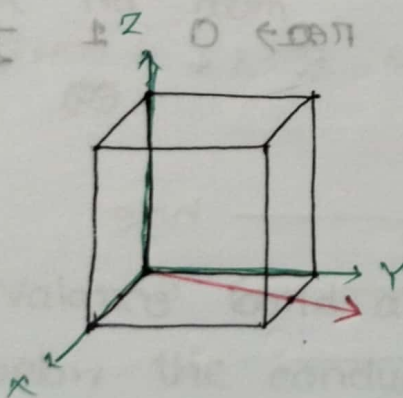
There are 8 corners of a unit cell. We can consider any corner with (x, y, z) axes in terms of the given value of (h, k, l)

* plane drawing এর জন্য reciprocal বের করতে highest integer দিয়ে ভাগ করতে হয়।

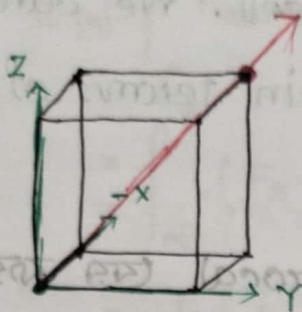
$[2 \ 1 \ 1]$



$[1 \ 2 \ 0]$



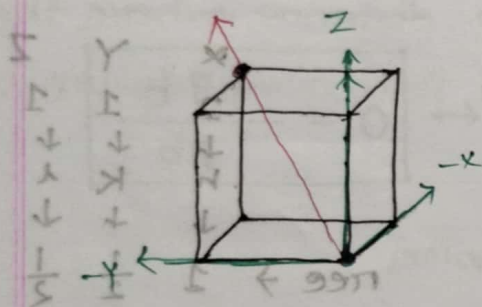
$$[\bar{1} \ 1 \ 1]$$



$$\begin{array}{ccc} \bar{1} & 1 & 1 \\ \downarrow & \downarrow & \downarrow \\ -1 & 1 & 1 \end{array}$$

$$\text{rec} \rightarrow -1 \ 1 \ 1$$

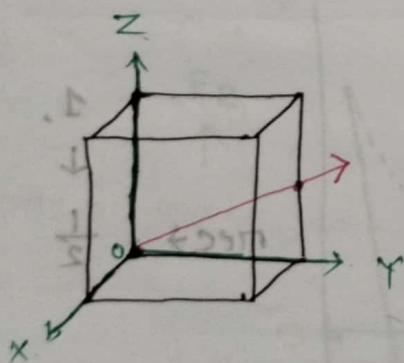
$$[\bar{1} \ \bar{1} \ 1]$$



$$\begin{array}{ccc} \bar{1} & \bar{1} & 1 \\ \downarrow & \downarrow & \downarrow \\ -1 & -1 & 1 \end{array}$$

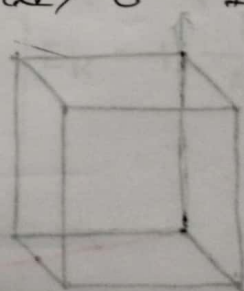
$$\text{rec} \rightarrow -1 \ -1 \ 1$$

$$[0 \ 2 \ 1]$$

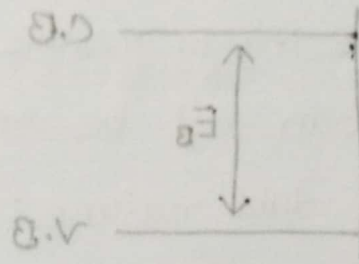
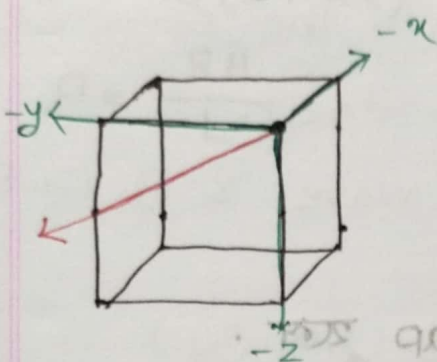


$$\begin{array}{ccc} 0 & 2 & 1 \\ \downarrow & \downarrow & \downarrow \\ 0 & 2 & 1 \end{array}$$

$$\text{rec} \rightarrow 0 \ 1 \ \frac{1}{2}$$

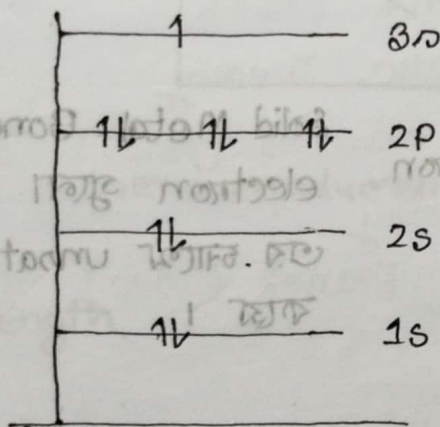


$[0 \ 2 \ 1]$

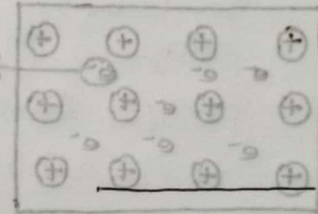


0	2	1
↓	↓	↓
0	-2	-1
0	-1	-1/2

Bond formation of solid:



$$M_s = \pm \frac{1}{2}$$



2 → high E

1.9

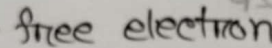
1.8 → low E

for Na atom

NB: যতগুলো atom একসাথে Bond করবে সেখানে E.L ও ততোটো ভাগ হবে।

- Valance bond are loosely bound electron. It is below the conduction band.
- Conduction band has free electron. It is also split in a seperate levels. Valance bond and conduction band has a energy level difference.

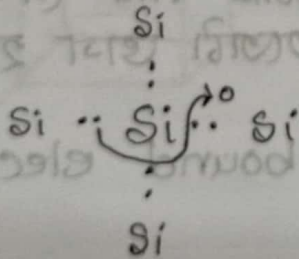
Metallic Bond :



Solid Metal Bond - এ free electron গুলো $+ve$ ion এর সাথে unsaturated Bond করে ।

- Conductor $\Rightarrow$ if $T \uparrow \Rightarrow R \downarrow \Rightarrow$ conductivity $\uparrow$

• Semi-conductor $\Rightarrow$ if $T \uparrow \Rightarrow R \downarrow \Rightarrow$ conductivity $\uparrow$



← hole

$$\xrightarrow{\text{valence } e^-}$$

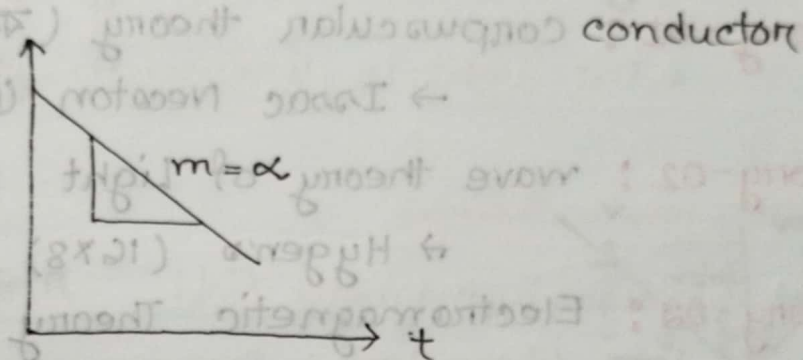
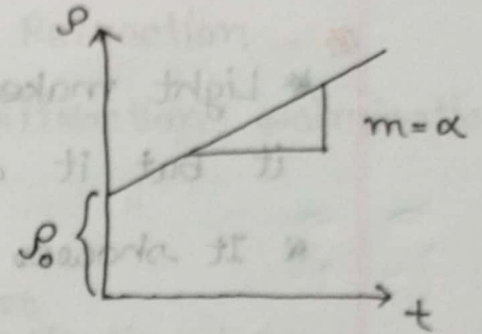
← Et wird

← I_e and I_n

Theory of Light

$$R = R_0(1 + \alpha t)$$

$$\rho = \frac{RA}{L}$$

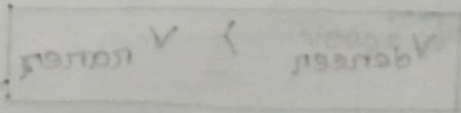


semi-conductor

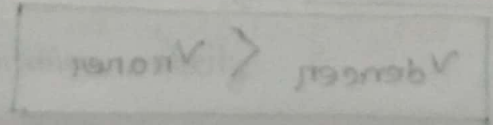
Bond-length

Theory of Light :-

Light emit from tiny elastic particles which are called corpuscles from luminous source of light.



but forward and backward proved.



Rectilinear propagation: The tendency of light to travel in a straight line.

Theory of Light

-Mamun Sir

- ☐ Light makes things visible with the presence of it but it can not be seen itself.
- ☐ It shows dual nature: ① wave & ② Particle.

Theory-01: corpuscular theory (কণাতত্ত্ব)

→ Isaac Newton (1672)

Theory-02: wave theory of Light (তরঙ্গতত্ত্ব)

→ Huygens (1678)

Theory-03: Electromagnetic Theory

→ J.C. Maxwell (1865)

Theory-04: Quantum Theory

→ Max Planck (1900)

Corpuscular Theory of Light :-

⇒ Light emit stream of tiny elastic particles which are called corpuscles from luminous source of light.
আলোকিত

$$\therefore v_{\text{denser}} > v_{\text{rarer}} \rightarrow \text{has limitation}$$

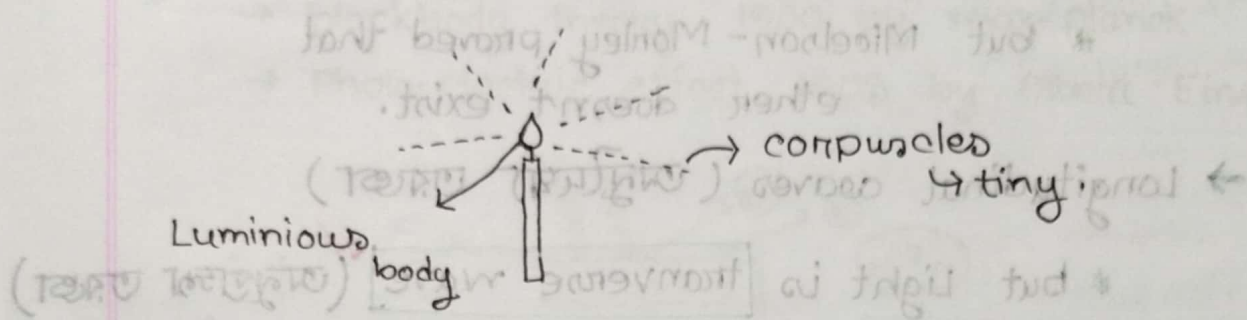
but Foucault and Michelson proved.

$$v_{\text{denser}} < v_{\text{rarer}} \quad \checkmark$$

→ Rectilinear Propagation: The tendency of light to travel in a straight line.

→ can explain : Reflection & Refraction

→ can't explain : Interference, Diffraction, polarization.

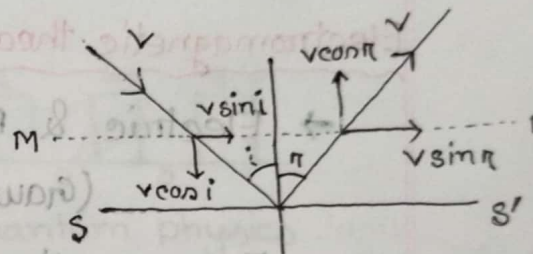


Reflection:

So, $v \sin i = v \sin r$

⇒ $i = r$

⇒ আপাতন কোণ = প্রতিফলন কোণ



Refraction:

$i > r$ & $v' < v$

Now, $v \sin i = v' \sin r$

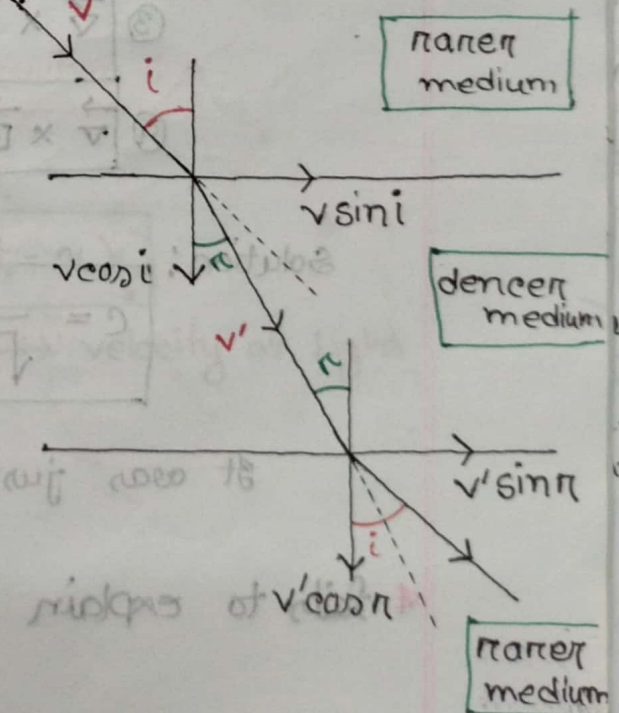
⇒ $\frac{v'}{v} = \frac{\sin i}{\sin r}$

again; $\mu = \frac{\sin i}{\sin r}$

$\mu = \frac{C_0}{C_m}$

C_0 → air medium velocity

C_m → any medium velocity



Wave theory:

→ Light travels as spherical waves through ether.

* but Michelson-Morley proved that ether doesn't exist.

→ Longitudinal waves (অনুদৈর্ঘ্য তরঙ্গ)

* but Light is transverse wave (অনুপ্রস্থ তরঙ্গ)

Electromagnetic theory:

→ Electric & magnetic field.

(Gauss Law + Ampere's Law + Faraday's Law)

Four equation:

$$\textcircled{1} \quad \vec{\nabla} \cdot \vec{E} = \frac{\rho}{\epsilon_0} \quad (\text{Gauss Law})$$

$$\textcircled{2} \quad \vec{\nabla} \cdot \vec{B} = 0 \quad (\text{Gauss Law})$$

$$\textcircled{3} \quad \vec{\nabla} \times \vec{E} = - \frac{\partial \vec{B}}{\partial t} \quad (\text{Faraday's law of electromagnetic induction})$$

$$\textcircled{4} \quad \vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \mu_0 \epsilon_0 \frac{\partial \vec{E}}{\partial t} \quad (\text{Ampere's Law})$$

Solution:

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$$

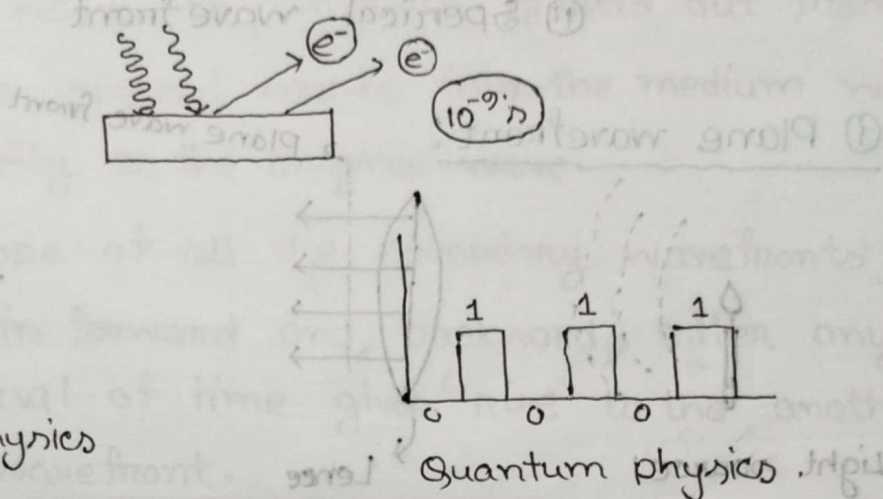
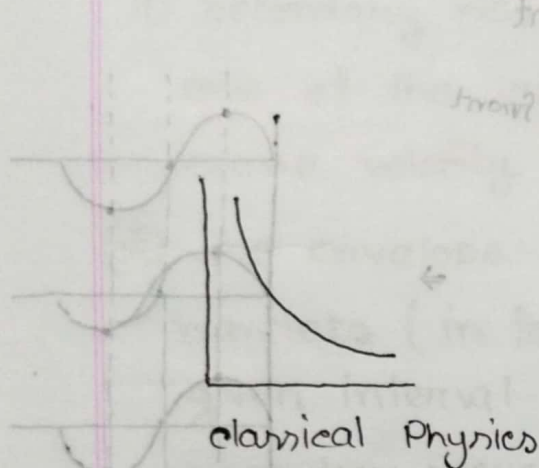
velocity of Light = $3 \times 10^8 \text{ ms}^{-1}$

It was justified in 1887 by F. Herz.

* fails to explain photo electric effect.

Quantum theory :-

- Lumps of Light → Quanta
- blackbody theory, 1900 by max-planck
- photo-electric effect, 1905 by Albert Einstein



- Bohr model, 1913
- De-Broglie Hypothesis, 1924
- uncertainty principle, 1927

Huygen's Principle and construction :

Equation of Progressive wave

$$y = a \sin \frac{2\pi}{\lambda} (ct - x)$$

→ velocity of Light

wave front:

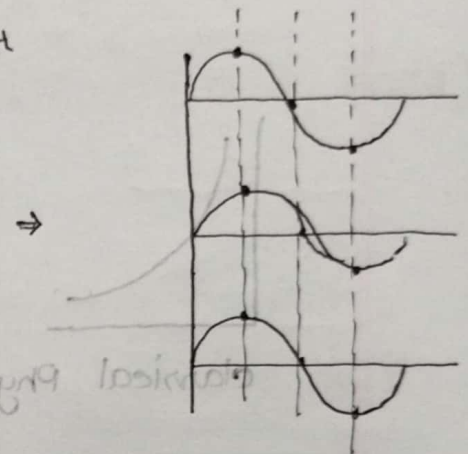
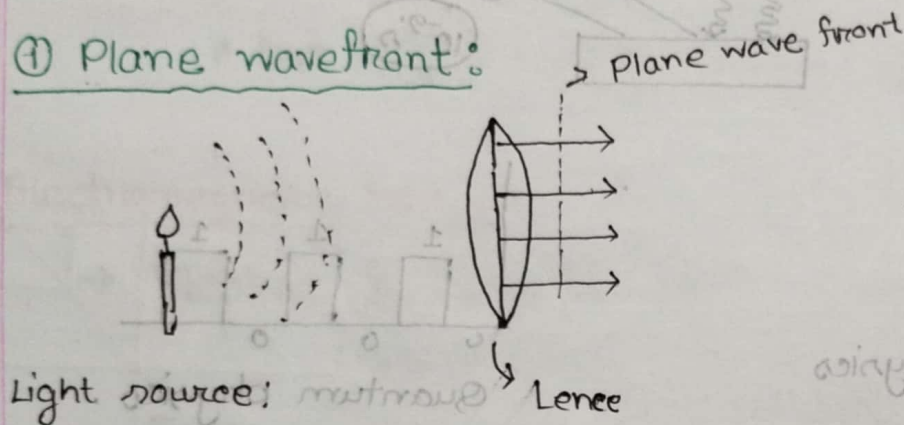
→ wave front is the path of particles having the same shape as the path of wave propagation:

2 types:

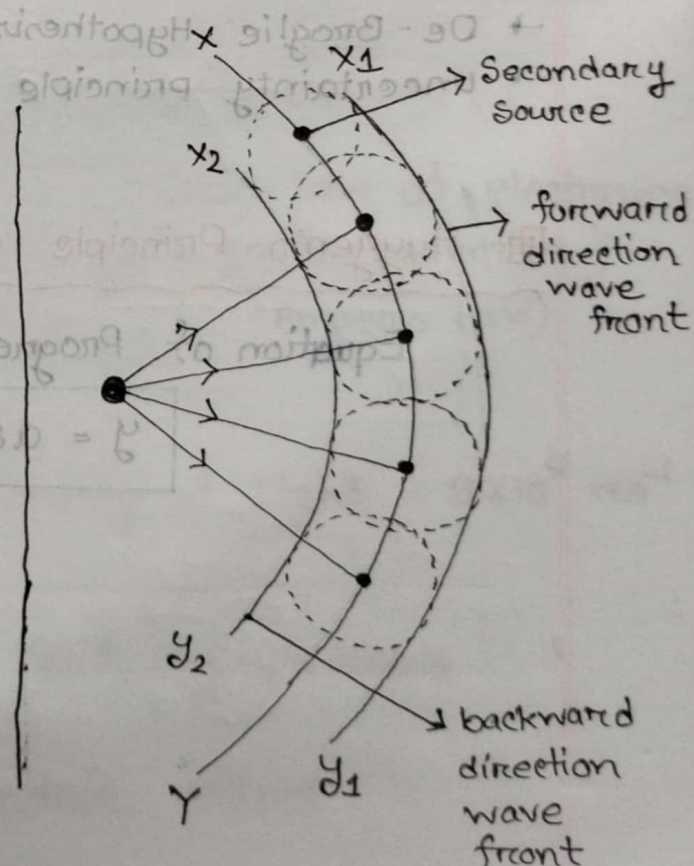
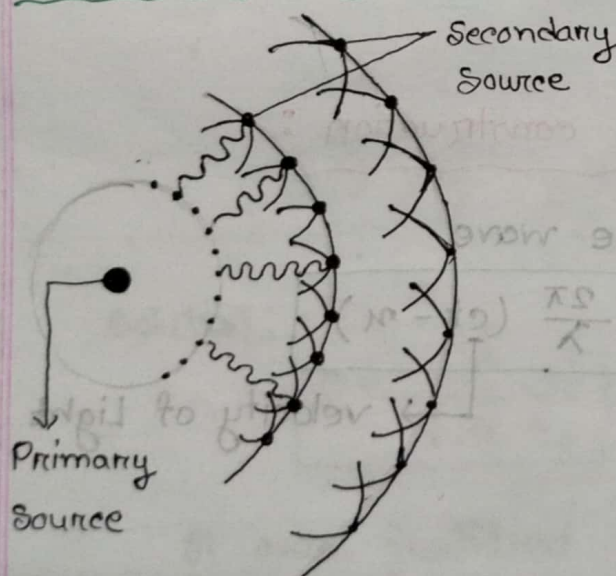
① Plane wavefront

② Spherical wavefront

① Plane wavefront:



② Spherical wavefront:



* Huygen's principle:- (spherical wavefront)

- i) Every point on a primary wavefront may be considered as a secondary source of disturbance.
- ii) Secondary waves or wavelets spread out from each one of the original source into the medium with the same velocity as the original wave.
- (iii) The envelope of all the secondary wavefronts or wavelets (in forward and backward) after any given interval of time gives rise to the another secondary wavefront.

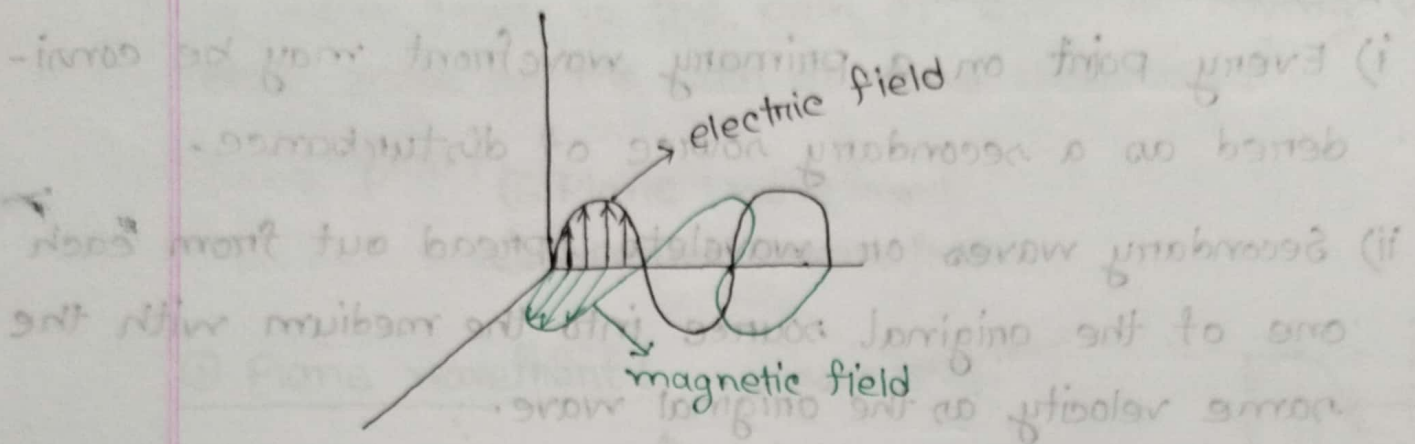
*** Q. Huygen's theory limitation & solve:

Drawback of Huygen's principle:-

- (i) It did not provide a clear mathematical framework for explaining complex wave phenomena. It can be overcome by Maxwell's eq. and electromagnetic theory.
- (ii) It shows difficulty in explaining interference and diffraction. It can be overcome by wave optics & wave theory.
- (iii) It can not explain the polarization of light. It can be overcome by polarization theory.
- (iv) Huygen's Principle did not address the particle like behaviour of light. It can be overcome by Quantum mechanics theory.

Solution:

* Electromagnetic theory (Huygens's principle) :-



Plane progressive wave eq. (iii)

$$y = a \sin \frac{2\pi}{\lambda} (\nu t - x)$$

$$\Rightarrow \frac{d^2 y}{dt^2} = \nu^2 \frac{\delta^2 y}{\delta x^2} \quad \text{--- (i)}$$

$$\nabla^2 E = \mu_0 \epsilon_0 \frac{\delta^2 E}{\delta t^2} \quad \text{--- (ii)}$$

$$\nabla^2 B = \mu_0 \epsilon_0 \frac{\delta^2 B}{\delta t^2} \quad \text{--- (iii)}$$

comparing these eq.

$$\nu^2 = \frac{1}{\mu_0 \epsilon_0}$$

$$\Rightarrow \nu = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = 3 \times 10^8$$

So, Light is a electromagnetic wave.

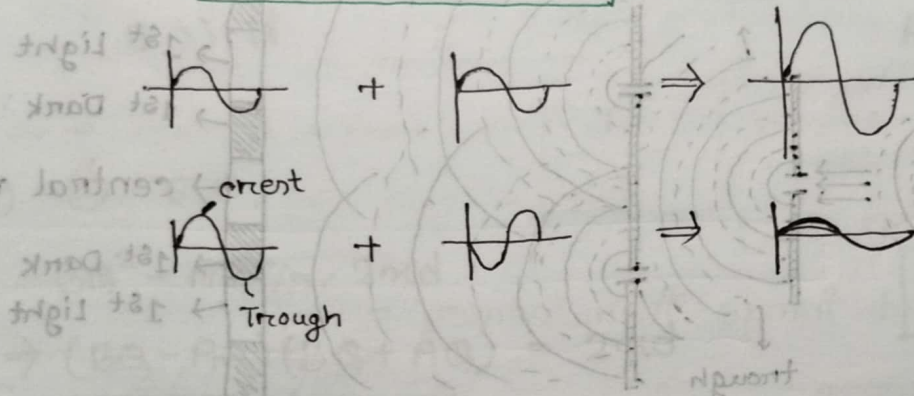
(Proved)

Interference :

Superposition theory of Light :-

When two or more waves of the same type (light waves) meet at a point in phase the resulting displacement at that point is the algebraic sum of the individual displacement of the waves.

$$\therefore y_{res} = y_1 + y_2$$



Interference :

→ The phenomena of multiple light waves interacting with one another under certain circumstances causing the combined amplitude of the waves to either increase or decrease.

Condition :

1. Amplitude and wavelength must be equal.
2. Distance between 2 source should be tends to zero.
3. Sources must be coherent (সুসঙ্গত)
4. Sources must be slim and thin.
(সূক্ষ্ম) (জন্ম)

wavelength increase $\rightarrow$ constructive case

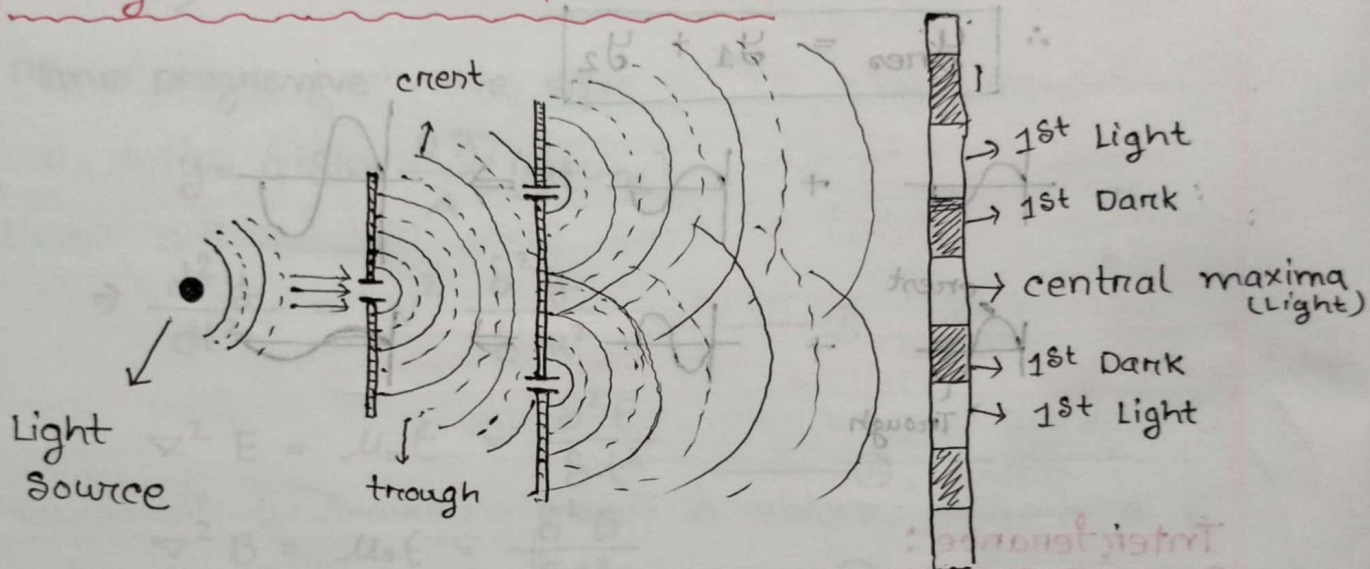
wavelength decrease $\rightarrow$ destructive case

$$\Delta \delta = \frac{2\pi}{\lambda} \cdot \Delta x$$

$\Delta \delta \rightarrow$ phase difference

$\Delta x \rightarrow$ path difference

Young's double slit experiment:



uneven distribution of light energy consisting of alternate light and dark spot.

constructive interference:-

same phase

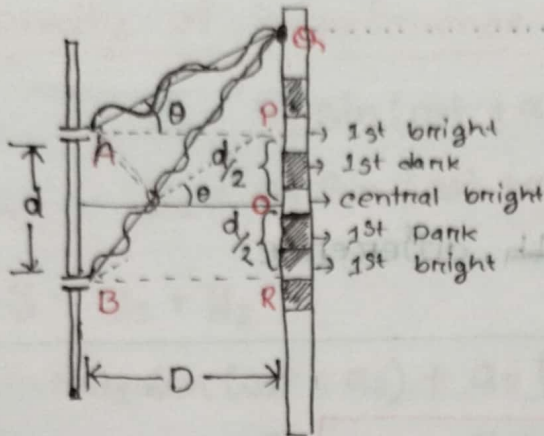
$$\Delta \delta = 0, 2\pi, 4\pi, 6\pi, \dots, 2n\pi$$

$$\Delta x = n\lambda = 2n \frac{\lambda}{2}$$

Path difference

even multiple of $\frac{\lambda}{2}$

Destructive interference:



Here

$$PQ = n - \frac{d}{2}$$

$$QR = n + \frac{d}{2}$$

$$\Delta BQR = 0$$

$$BQ^2 = BR^2 + QR^2$$

$$= D^2 + \left(n + \frac{d}{2}\right)^2 \quad \text{--- (i)}$$

$$\Delta APQ = 0$$

$$AQ^2 = AP^2 + PQ^2$$

$$= D^2 + \left(n - \frac{d}{2}\right)^2 \quad \text{--- (ii)}$$

(i) - (ii) से,

$$BQ^2 - AQ^2 = 2nd$$

$$\Rightarrow (BQ - AQ)(BQ + AQ) = 2nd$$

$$\Rightarrow BQ - AQ = \frac{2nd}{2D}$$

$$\Rightarrow n\lambda = \frac{nd}{D}$$

$$\Rightarrow n_n = \frac{n\lambda D}{d}$$

$$\Rightarrow (2n+1)\frac{\lambda}{2} = \frac{nd}{D}$$

$$\Rightarrow n_n = \frac{(2n+1)\lambda D}{2d}$$

$$BQ + AQ \approx 2D$$

$$BQ - AQ = \Delta x$$

for bright fringe

$$\Delta x = n\lambda$$

for dark fringe

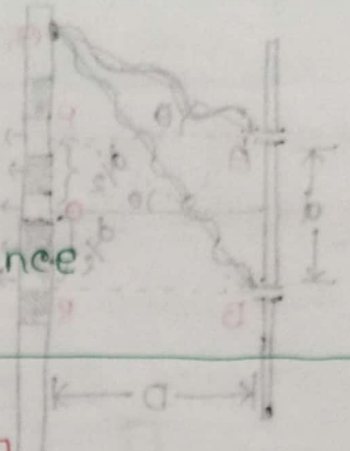
$$\Delta x = (2n+1)\frac{\lambda}{2}$$

The difference between two bright / dark fringe:

$$x_2 - x_1 = \frac{\lambda D}{d}$$

$$\Rightarrow \boxed{\beta = \frac{\lambda D}{d}}$$

fringe width, difference



Overall Equations:

* Fringe difference $\Delta \beta = \frac{\lambda D}{d}$, fringe width

* Fringe ~~width~~ ^{breadth} = $\frac{\lambda D}{2a}$ The breadth of dark on bright
(জোরা প্রস্থ)

* nth bright fringe from center = $\frac{n \lambda D}{d}$

* nth dark fringe from center = $\frac{(2n+1) \lambda D}{2d}$

* angular diff of fringe, $\tan \theta_n = \frac{x_n}{D}$, $\sin \theta_n = \frac{\Delta x}{d}$

* Max bright fringe $d \sin \theta_n = n \lambda$

* max dark fringe $d \sin \theta_n = (2n+1) \frac{\lambda}{2}$

* for good interference $\beta \propto D$

স্ক্রীন দূরত্ব বারোতে হবে,

অথবা, $\boxed{\beta \propto \frac{1}{d}}$

দুই slit ৯ পরস্পর দূরত্ব কমতে হবে

Thin film interference:

Intensity of interference:-

$$y_1 = a_1 \sin(\omega t + \alpha_1)$$

$$y_2 = a_2 \sin(\omega t + \alpha_2)$$

$$\therefore y = y_1 + y_2$$

$$= a_1 \sin(\omega t + \alpha_1) + a_2 \sin(\omega t + \alpha_2)$$

$$= a_1 \sin \omega t \cos \alpha_1 + a_1 \sin \alpha_1 \cos \omega t + a_2 \sin \omega t \cos \alpha_2 + a_2 \sin \alpha_2 \cos \omega t$$

$$= \sin \omega t (a_1 \cos \alpha_1 + a_2 \cos \alpha_2) + \cos \omega t (a_1 \sin \alpha_1 + a_2 \sin \alpha_2)$$

$$= \sin \omega t \cdot A \cos \phi + \cos \omega t \cdot A \sin \phi$$

$$= A \sin(\omega t + \phi)$$

$$\text{Now, } A \cos \phi = a_1 \cos \alpha_1 + a_2 \cos \alpha_2 \quad \text{--- (i)}$$

$$A \sin \phi = a_1 \sin \alpha_1 + a_2 \sin \alpha_2 \quad \text{--- (ii)}$$

$$\text{(i)}^2 + \text{(ii)}^2 \Rightarrow$$

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos(\alpha_1 - \alpha_2)$$

→ Grained amplitude.

$$\text{(ii)} \div \text{(i)} \Rightarrow$$

$$\tan \theta = \frac{a_1 \sin \alpha_1 + a_2 \sin \alpha_2}{a_1 \cos \alpha_1 + a_2 \cos \alpha_2}$$

→ Grained Phase.

We know;

$$I \propto a^2$$

$$A^2 = a_1^2 + a_2^2 + 2a_1a_2 \cos \Delta \delta$$

$$\therefore I = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos \Delta \delta \quad \rightarrow \text{Intensity.}$$

But in Interference,

$$I_1 = I_2 = I_0$$

$$\therefore I = 2I_0 + 2I_0 \cos \Delta \delta$$

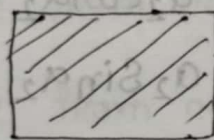
$$= 2I_0 (1 + \cos \Delta \delta)$$

$$= 2I_0 \cdot 2 \cos^2 \frac{\Delta \delta}{2}$$

$$\therefore I = 4I_0 \cos^2 \frac{\Delta \delta}{2} \rightarrow \text{Intensity in Interference.}$$

Thin film interference:-

A layer of transparent material with thickness ranging from nm to Å.



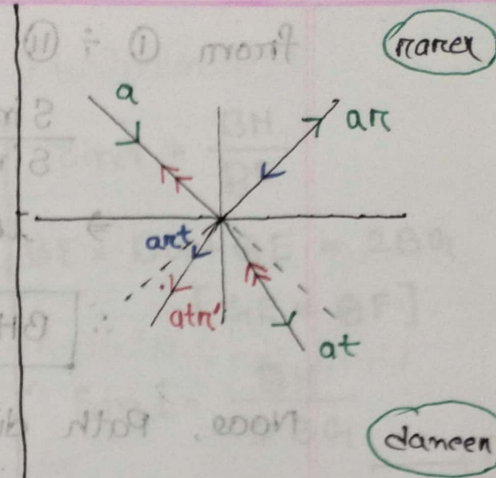
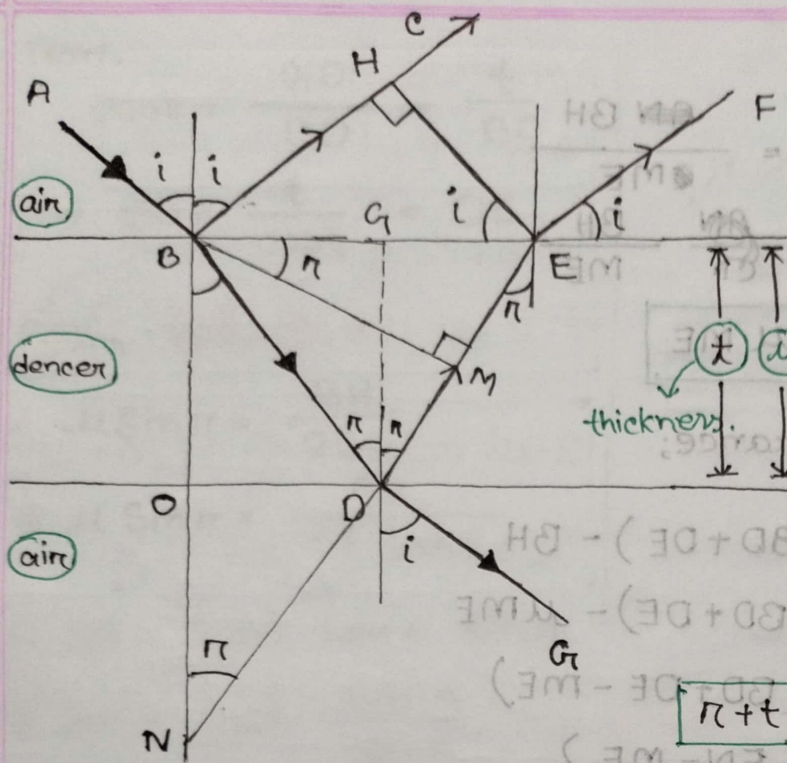
Thin film interference -

(i) reflections

ii) Transmission

The difference in the distance traveled by 2 Light rays or waves as they pass through different paths in a medium optical system is called optical path difference.

$$\text{optical path} = \mu \times \text{Geometrical path}$$



$$\begin{aligned} \pi t + \pi n' &= 0 \\ \Rightarrow \pi + \pi' &= 0 \\ \Rightarrow \pi &= -\pi' \end{aligned}$$

$$\pi + t = 1$$

$$\begin{aligned} \text{Phase diff} &= \frac{2\pi}{\lambda} \times \text{path diff} \\ \Rightarrow \Delta \phi &= \frac{2\pi}{\lambda} \times \Delta x \end{aligned}$$

Now,

optical path diff between BC & EF

$$\Delta x = [\mu (BD + DE) - BH] - \frac{\lambda}{2}$$

if phase change to π
তখন হলে যখন phase পরিবর্তন করা হবে।
না হলে, 0।

Now,

$$\cos \pi = \frac{GD}{BD} = \frac{t}{BD}$$

$$\Rightarrow BD = \frac{t}{\cos \pi} = DE$$

$$\mu \sin \pi = \frac{BH}{2BG}$$

$$\Rightarrow \mu \sin \pi = \frac{BH}{2t \tan \pi}$$

$$\Rightarrow BH = 2\mu t \tan \pi \sin \pi$$

$$\Rightarrow BH = 2\mu t \frac{\sin^2 \pi}{\cos \pi}$$

Now,

path difference,

$$\Delta x = [\mu(BD + DE) - BH] \cdot \frac{\lambda}{2}$$

$$= \left(\frac{2\mu t}{\cos \pi} - 2\mu t \frac{\sin^2 \pi}{\cos \pi} \right) \cdot \frac{\lambda}{2}$$

$$\Rightarrow \frac{2\mu t (1 - \sin^2 \pi)}{\cos \pi} \cdot \frac{\lambda}{2}$$

$$\Delta x = 2\mu t \cos \pi \cdot \frac{\lambda}{2}$$

for transmitter ray $\Delta x = 2\mu t \cos \pi$

$\Delta x = n\lambda \rightarrow$ Bright

$\Delta x = (2n+1) \frac{\lambda}{2} \rightarrow$ Dark

Now, determine

$$\sin i = \frac{BH}{BE}$$

$$BE = BG + GE = 2BG$$

$$[GE = BF]$$

$$\therefore \sin i = \frac{BH}{2BG}$$

Snell's law

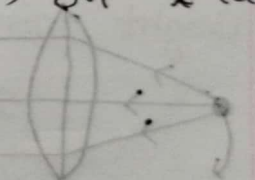
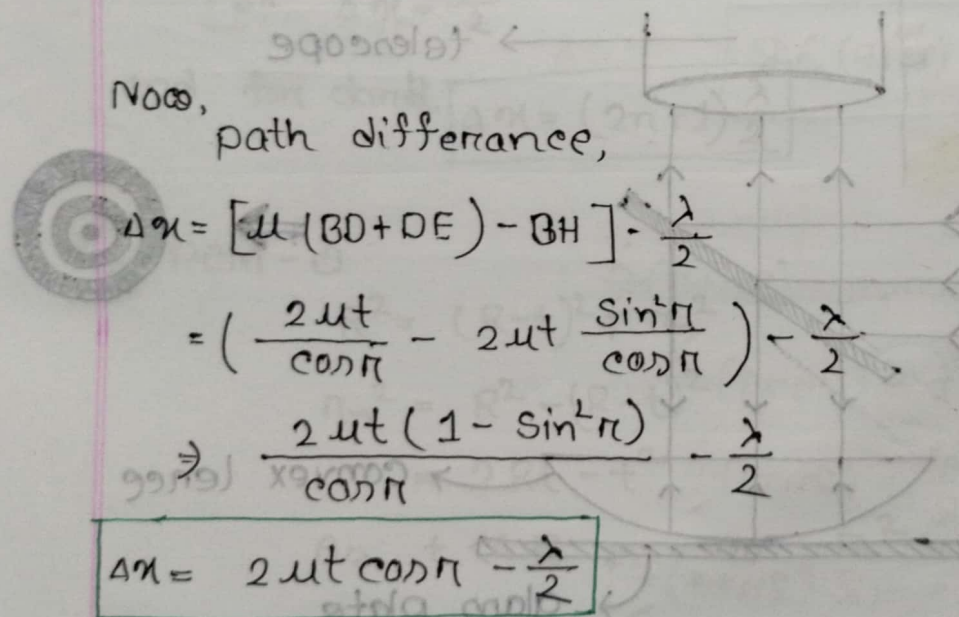
$$\mu = \frac{\sin i}{\sin \pi}$$

$$\Rightarrow \mu \sin \pi = \sin i$$

$$\tan \pi = \frac{BG}{GD}$$

$$\Rightarrow \tan \pi = \frac{BG}{t}$$

$$\Rightarrow BG = t \tan \pi$$



Transmitted ray,

$$\Delta x = 2 \mu t \cos r$$

Bright $\leftarrow 2 \mu t \cos r = n \lambda$

Dark $\leftarrow 2 \mu t \cos r = (2n+1) \frac{\lambda}{2}$

reflected ray,

$$\Delta x = 2 \mu t \cos r - \frac{\lambda}{2}$$

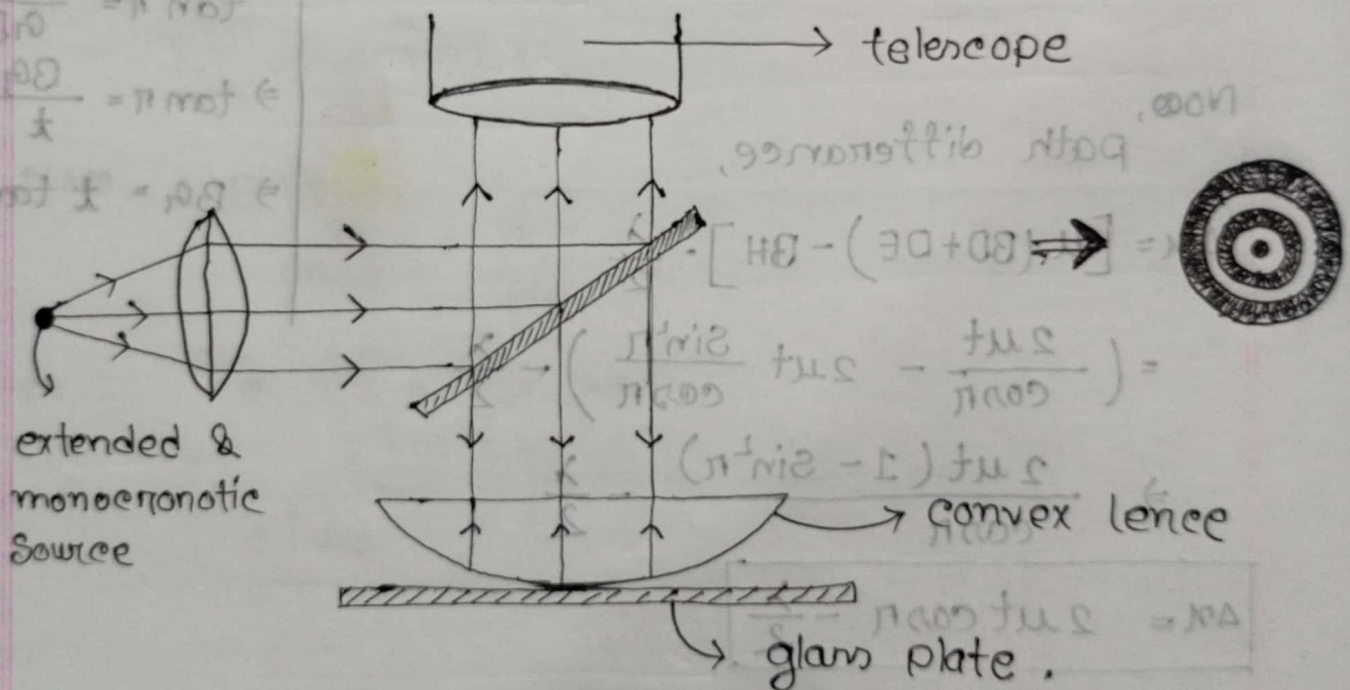
Bright $\rightarrow 2 \mu t \cos r - \frac{\lambda}{2} = n \lambda$

$$\Rightarrow 2 \mu t \cos r = (2n+1) \frac{\lambda}{2}$$

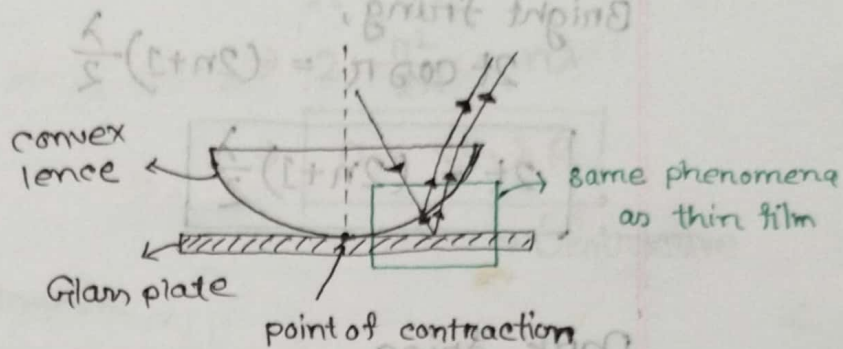
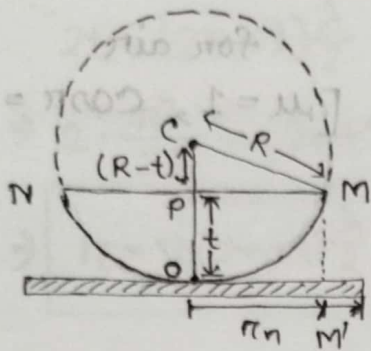
Dark $\rightarrow 2 \mu t \cos r - \frac{\lambda}{2} = (2n+1) \frac{\lambda}{2}$

$$\Rightarrow 2 \mu t \cos r = (n+1) \lambda$$

Newton's Ring Experiment



Newton's rings are circular in shape as the air film is symmetrical about the point of contact of the lens with the glass plate.



* why is the central ring dark,

$$\Delta x = 2 \mu t \cos \theta \pm \frac{\lambda}{2}$$

at point of contact $t = 0$

$$\text{So, } \Delta x = \frac{\lambda}{2}$$

and for dark, $\Delta x = (2n+1) \frac{\lambda}{2}$

$$OC = R = CM$$

$$OP = t$$

$$PC = OC - OP = R - t$$

$$OM' = r_n \text{ (গোলক / লেন্সের ছায়ার ব্যাসার্ধ)}$$

$$\Delta PCM - \Delta$$

$$R^2 = (R-t)^2 + r_n^2$$

$$r_n^2 = R^2 - (R-t)^2$$

$$= 2Rt - t^2$$

$$\text{As } t \ll R, \text{ so } r_n^2 = 2Rt \Rightarrow t = \frac{r_n^2}{2R}$$

$$r_n = \text{radius of observed ring.}$$

$\mu = 1$, for air,
Bright fring,

$$2t \cos \pi = (2n+1) \frac{\lambda}{2}$$

$$2t = (2n+1) \frac{\lambda}{2}$$

Dark fring,

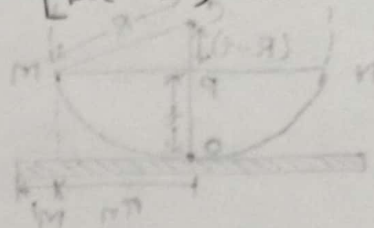
$$2\mu t \cos \pi = m \lambda$$

$$\Rightarrow 2 \times \frac{D_m^2}{8R} = m \lambda$$

$$\Rightarrow D_m^2 = 4m \lambda R$$

$$\Rightarrow D_{m+p}^2 = 4(m+p) \lambda R$$

for air
[$\mu = 1$, $\cos \pi = 1$]



$$\frac{D_m^2}{4} = \frac{D_m^2}{4}$$

$$t = \frac{D_m^2}{8R}$$

$$m = (n+1)$$

Bright fring,

$$2\mu t \cos \pi = (2n+1) \frac{\lambda}{2}$$

$$\Rightarrow 2t = (2n+1) \frac{\lambda}{2}$$

$$\Rightarrow \frac{D_n^2}{4R} = (2n+1) \frac{\lambda}{2}$$

$$\Rightarrow D_n^2 = 4R (2n+1) \frac{\lambda}{2}$$

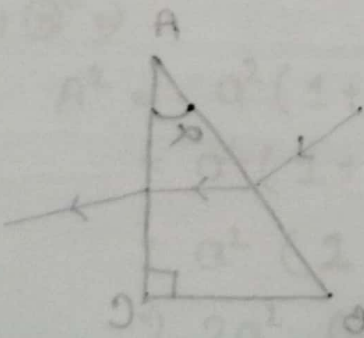
$$\Rightarrow D_{n+p}^2 = 4R (2(n+p)+1) \frac{\lambda}{2}$$

extra:

$$\begin{aligned}
 2t &= (2n+1) \frac{\lambda}{2} & \cdot 2t &= n\lambda \\
 \Rightarrow 2 \cdot \frac{r^2}{2R} &= (2n+1) \frac{\lambda}{2} & \Rightarrow 2 \cdot \frac{r^2}{2R} &= n\lambda \\
 \Rightarrow \boxed{r = \sqrt{(2n+1) \frac{\lambda}{2} \cdot R}} & & \Rightarrow \boxed{r = \sqrt{n\lambda R}} & \text{Destructive} \\
 & \text{constructive} & &
 \end{aligned}$$

Ring difference:

$$\begin{aligned}
 r_n &= \sqrt{n\lambda R} & \in (1) - (4) \\
 \Rightarrow D_n^2 &= 4n\lambda R & \text{--- (1)} \\
 \Rightarrow D_{n+p}^2 &= 4(n+p)\lambda R & \text{--- (2)} \\
 (2) - (1) & \Rightarrow D_{n+p}^2 - D_n^2 = 4(n+p-n)\lambda R \\
 \Rightarrow \boxed{D_{n+p}^2 - D_n^2 = 4p\lambda R} \\
 \Rightarrow \boxed{\lambda = \frac{D_{n+p}^2 - D_n^2}{4pR}}
 \end{aligned}$$



Here, AB and AC are the sides of the triangle and BC is the base. If $\alpha > 90^\circ$, then the prism is called an obtuse prism.

Newton's ring experiment in different field:-

Now, refraction factor, $\mu > 1$ (say Liquid)

$$\text{So, } D_{n_1}^2 = \frac{4n\lambda R}{\mu} \quad \text{--- (i)}$$

$$D_{(n+p)_1}^2 = \frac{4(n+p)\lambda R}{\mu} \quad \text{--- (ii)}$$

$$\text{(i) - (ii) } \Rightarrow$$

$$D_{(n+p)_1}^2 - D_{n_1}^2 = \frac{4p\lambda R}{\mu}$$

$$\Rightarrow [D_{(n+p)}^2 - D_n^2]_{\text{liquid}} = \frac{[D_{(n+p)}^2 - D_n^2]_{\text{air}}}{\mu}$$

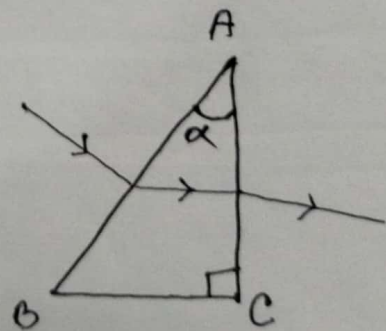
$$\Rightarrow \mu = \frac{[D_{(n+p)}^2 - D_n^2]_{\text{air}}}{[D_{(n+p)}^2 - D_n^2]_{\text{liquid}}} = \frac{4p\lambda R}{D_{n+p}^2 - D_n^2} \quad \text{--- (iii)}$$

extra:

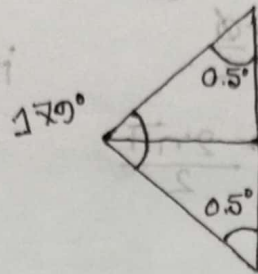
Here, AB and AC refracting surface & BC = Base

if $\alpha < 90^\circ$, then

the prism is called acute angle of prism.



Fresnel by-prism : $\Rightarrow$ $\frac{1}{2}$ light $\frac{1}{2}$!



Intensity of interference (Part-2)

$$y_1 = a \sin \omega t, \quad y_2 = a \sin (\omega t + \phi)$$

Now,

$$y = y_1 + y_2$$

$$= a \sin \omega t + (a \sin \omega t + \phi)$$

$$= a \sin \omega t + a \sin \omega t \cos \phi + a \sin \phi \cos \omega t$$

$$\Rightarrow (a + a \cos \phi) \sin \omega t + a \sin \phi \cos \omega t$$

$$\Rightarrow A \cos \theta \sin \omega t + A \sin \theta \cos \omega t$$

$$\Rightarrow A \sin(\omega t + \theta)$$

Here,

$$a + a \cos \phi = A \cos \theta \quad \text{--- (i)}$$

$$a \sin \phi = A \sin \theta \quad \text{--- (ii)}$$

$$\text{(i)}^2 + \text{(ii)}^2 \Rightarrow$$

$$A^2 = a^2 (1 + \cos \phi)^2 + a^2 \sin^2 \phi$$

$$= a^2 (1 + 2 \cos \phi + \cos^2 \phi + \sin^2 \phi)$$

$$= a^2 (1 + 2 \cos \phi)$$

$$\Rightarrow 2a^2 (1 + \cos \phi)$$

$$\Rightarrow 2a^2 \cdot 2 \cos^2 \frac{\phi}{2}$$

$$A^2 \Rightarrow 4a^2 \cos^2 \frac{\phi}{2}$$

we know,

$$I \propto A^2$$

if constant = 1

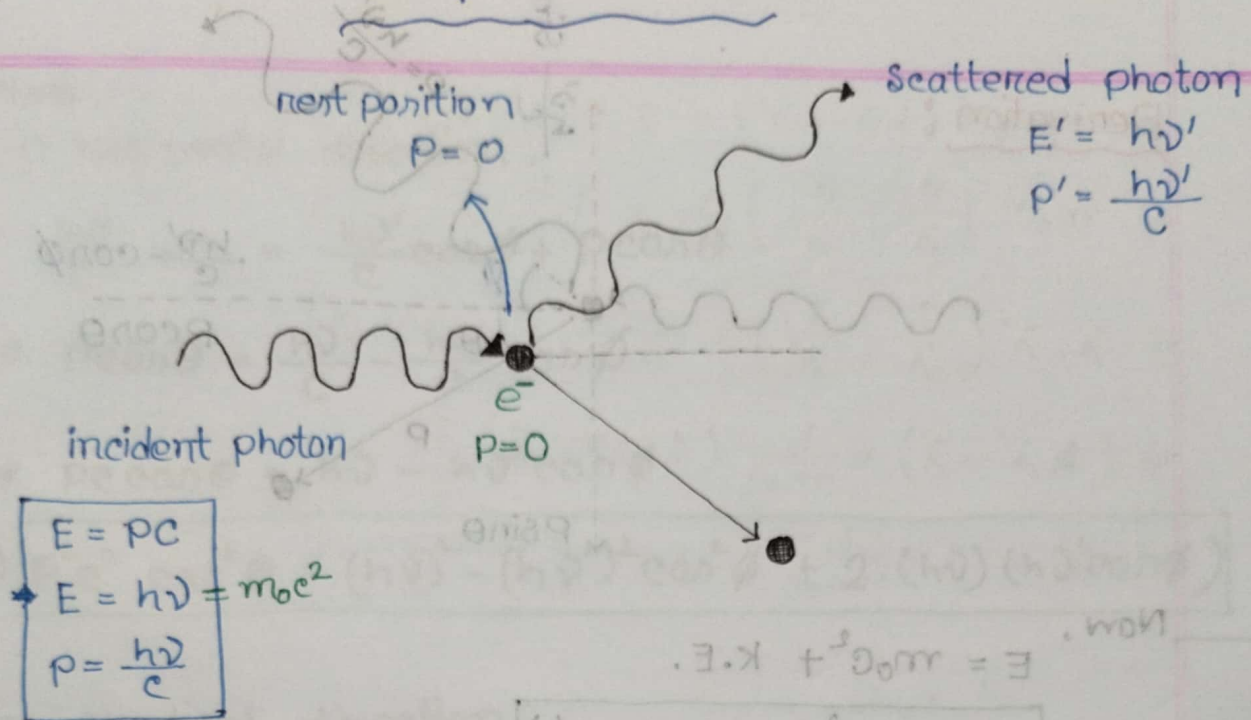
$$\Rightarrow I = 4a^2 \cos^2 \frac{\phi}{2}$$

if $\phi = 2n\pi$

$$\Rightarrow I = 4a^2 \cos^2 \frac{2n\pi}{2}$$

$$\Rightarrow \boxed{I = 4a^2}$$

Compton Effect



$$\text{①} \quad E = m_0 c^2 + K.E.$$

Now, Loss of photon energy = gain in K.E. of electron.

$$\Rightarrow E - E' = h\nu - h\nu' = K.E.$$

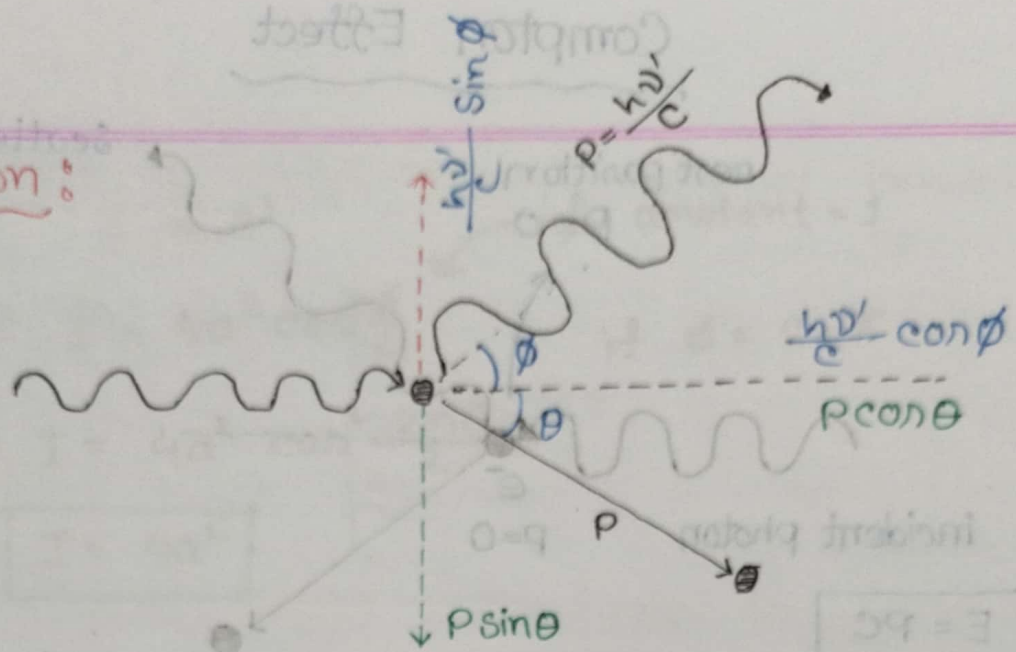
From theory of relativity of mass we know,

$$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$$

$$\Rightarrow m_0 c^2 = \sqrt{p^2 c^2 + m_0^2 c^4}$$

$$\Rightarrow E = \sqrt{p^2 c^2 + m_0^2 c^4}$$

Derivation:



Now,

$$E = m_0 c^2 + K.E.$$

$$\Rightarrow E = m_0 c^2 + (h\nu - h\nu') \quad \text{--- (i)}$$

again, theory of R. M

$$E = \sqrt{p^2 c^2 + m_0^2 c^4} \quad \text{--- (ii)}$$

(i) & (ii) $\Rightarrow$

$$\sqrt{p^2 c^2 + m_0^2 c^4} = m_0 c^2 + (h\nu - h\nu')$$

$$\Rightarrow p^2 c^2 + m_0^2 c^4 = m_0^2 c^4 + (h\nu - h\nu')^2 + 2 \cdot m_0 c^2 \cdot (h\nu - h\nu')$$

$$\Rightarrow p^2 c^2 = h^2 \nu^2 - 2 h\nu \cdot h\nu' + h^2 \nu'^2 + 2 m_0 c^2 \cdot (h\nu - h\nu') \quad \text{--- (iii)}$$

Now, in horizontal direction,

$$\frac{h\nu}{c} + 0 = \frac{h\nu'}{c} \cos \phi + p \cos \theta$$

$$\Rightarrow p \cos \theta = \frac{h\nu}{c} - \frac{h\nu'}{c} \cos \phi$$

$$\Rightarrow p \cos \theta = h\nu - h\nu' \cos \phi$$

$$\Rightarrow p^2 \cos^2 \theta = (h\nu)^2 - (h\nu')^2 \cos^2 \phi + 2(h\nu)(h\nu' \cos \phi)$$

(iv)

in vertical direction,

$$0 + 0 = \frac{h\nu'}{c} \sin \phi - p \sin \theta$$

$$\Rightarrow p \sin \theta = h\nu' \sin \phi$$

$$\Rightarrow p^2 \sin^2 \theta = (h\nu')^2 \sin^2 \phi$$

(v)

$$(iv) + (v) \Rightarrow$$

$$p^2 c^2 = (h\nu)^2 + (h\nu')^2 - 2(h\nu)(h\nu') \cos \phi$$

(vi)

Now, from (iii) & (vi) $\Rightarrow$

$$-2(h\nu)(h\nu') \cos \phi = -2(h\nu)(h\nu') + 2m_0 c^2 (h\nu - h\nu')$$

$$\Rightarrow 2(h\nu)(h\nu') - 2(h\nu)(h\nu') \cos \phi = 2m_0 c^2 (h\nu - h\nu')$$

$$\Rightarrow 2 m_0 c^2 (h\nu - h\nu') = 2 (h\nu)(h\nu') (1 - \cos\phi)$$

$$\Rightarrow m_0 c^2 \left(\frac{c}{\lambda} - \frac{c}{\lambda'} \right) \cdot h = h^2 \frac{c^2}{\lambda \cdot \lambda'} (1 - \cos\phi)$$

$$\Rightarrow \lambda \cdot \lambda' \left(\frac{1}{\lambda} - \frac{1}{\lambda'} \right) = \frac{h}{m_0 c} (1 - \cos\phi)$$

$$\Rightarrow (\lambda' - \lambda) = \frac{h}{m_0 c} (1 - \cos\phi)$$

$$\Rightarrow \lambda' = \lambda + \frac{h}{m_0 c} (1 - \cos\phi)$$

$$\text{So, } \lambda' > \lambda$$

So, scattered ray is greater than incident ray.

$$\Delta \lambda = \frac{h}{m_0 c} (1 - \cos\phi)$$

$\Delta \lambda$ will be maximum, when $\phi = 180^\circ$

$$\text{then, } \Delta \lambda_{\max} = 0.048 \text{ \AA}$$

*** why x-ray shows observed Compton effect?

(iv)

← (iv) & (iii) must, work

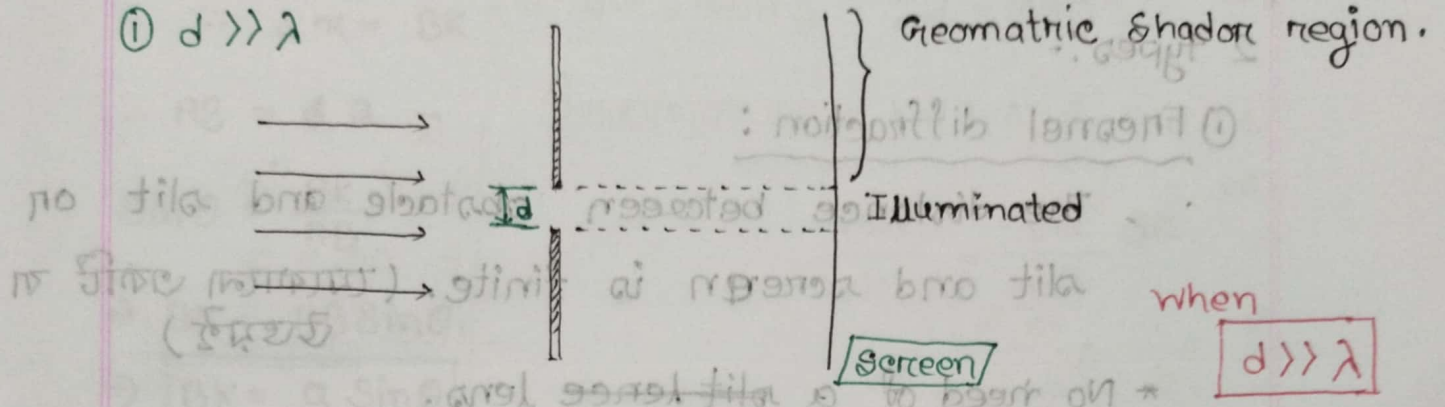
$$(h\nu - h\nu')^2 = 2 m_0^2 c^4 + 2 h\nu h\nu' (1 - \cos\phi) = 2 m_0^2 c^4 + 2 m_0^2 c^4 \frac{h\nu h\nu'}{m_0^2 c^4} (1 - \cos\phi)$$

$$(h\nu - h\nu')^2 = 2 m_0^2 c^4 + 2 m_0^2 c^4 \frac{h\nu h\nu'}{m_0^2 c^4} (1 - \cos\phi)$$

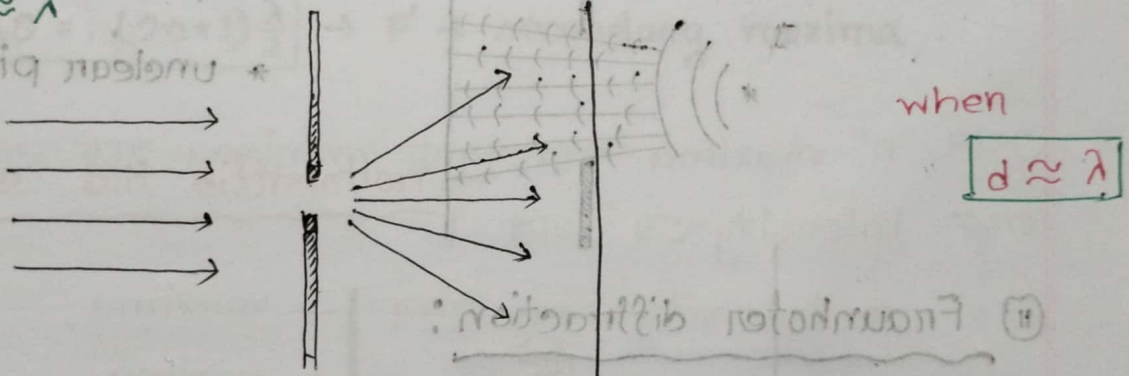
Diffraction of Light

two condition :-

① $d \gg \lambda$



② $d \approx \lambda$



The slight bending of light near the edges of an obstacle or opening or slit and spreading into the region of geometric shadows, this phenomena is called diffraction.

when

$\lambda > d \rightarrow$ আলো বোঁকি ছড়ায়, জগমগে প্যাটার্ন, slit এর কাছে diff. হয়।

$\lambda < d \rightarrow$ diffraction দুর্বল; আলো শ্রায় জোড়া যায়, দূরে গিয়া diff. বোঝা যায়।

$\lambda \approx d \rightarrow$ সবচেয়ে জটিল diff. প্যাটার্ন দেখা যায়, screen এ বোঝা যায়।

Diffraction of light

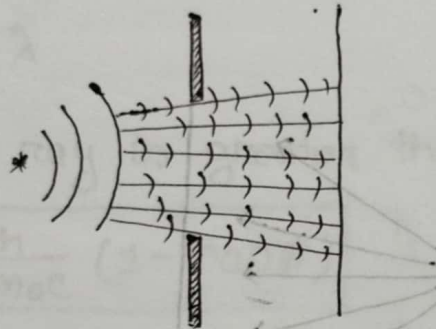
Types of diffraction:-

2 types:-

(i) Fresnel diffraction:

The distance between obstacle and slit or slit and screen is finite. (যেকোনো একটি বা উভয়ই)

* No need of a ~~slit~~ lens.



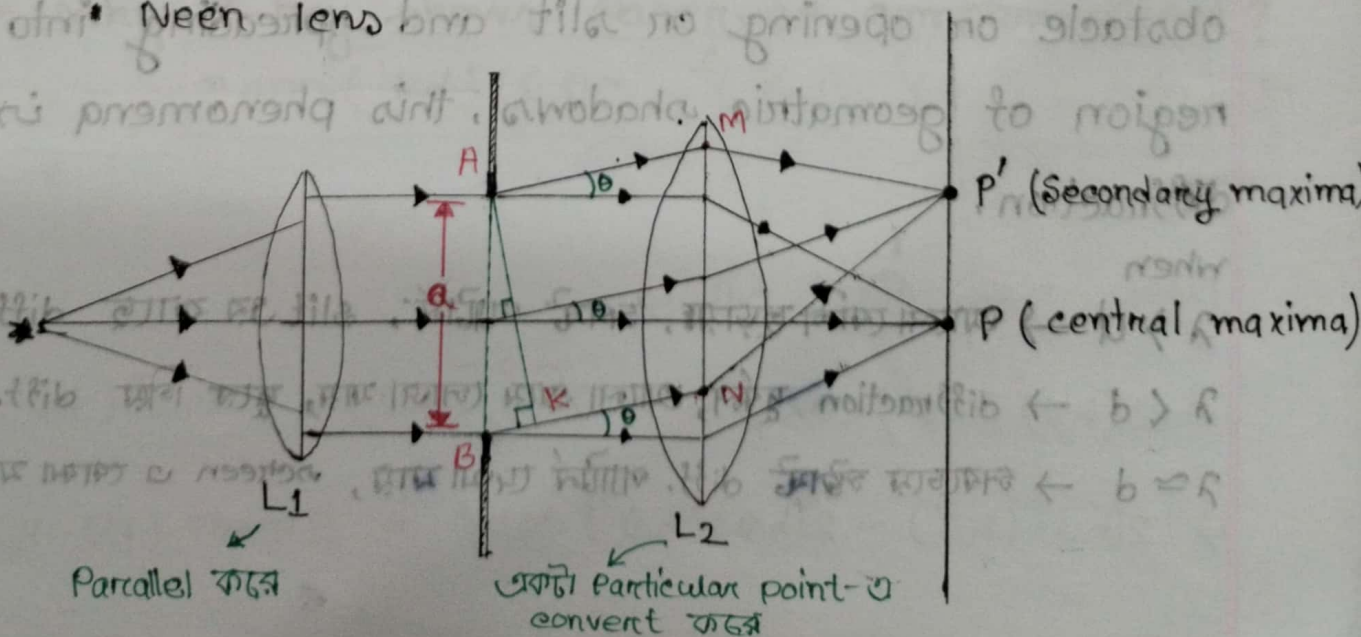
* spherical wavefront.

* unclear picture.

(ii) Fraunhofer diffraction:

The distance between obstacle and slit or slit and screen is infinite.

* Need lens



Path diff of AM & BN = BK

$$\therefore \Delta x = BK$$

$$\therefore AB = a$$

$$\sin \theta = \frac{BK}{AB}$$

$$\Rightarrow BK = AB \sin \theta$$

$$\Rightarrow BK = a \sin \theta$$

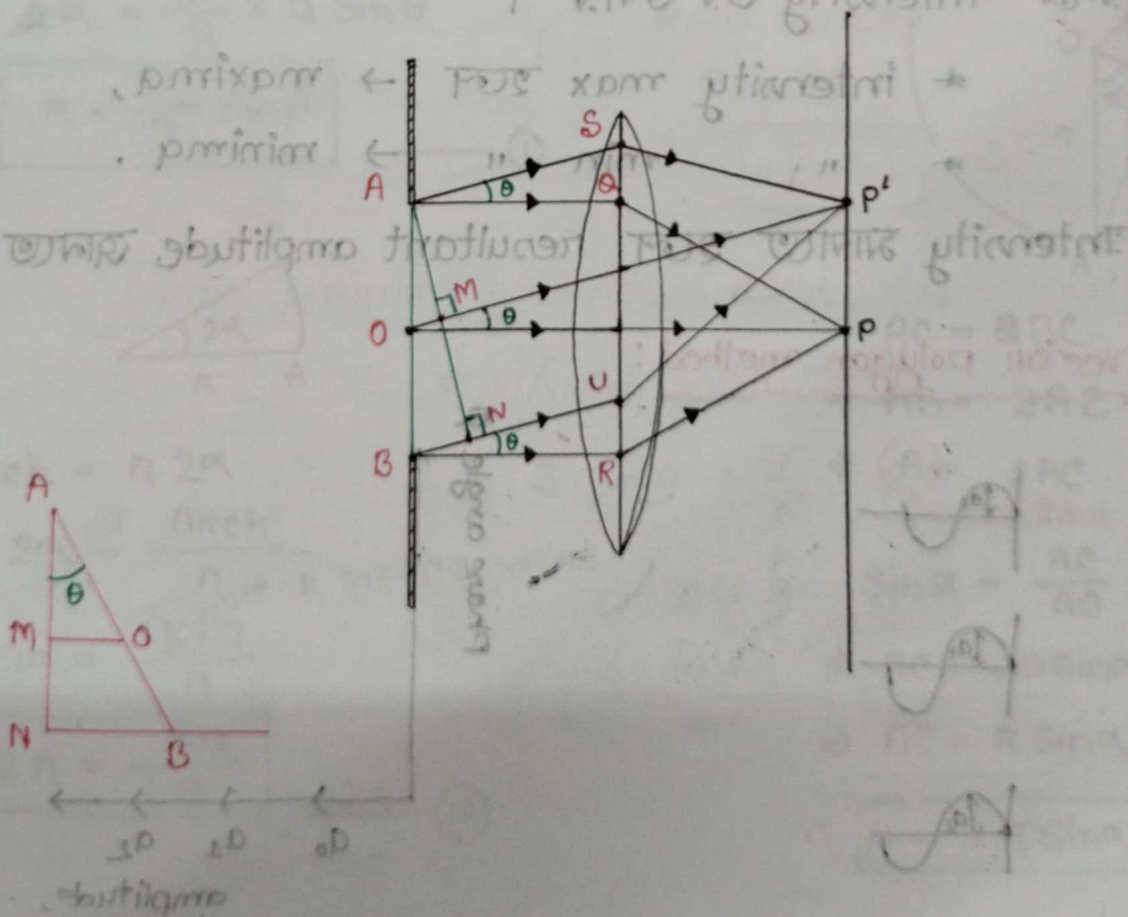
$$a \sin \theta = n \lambda$$

$\rightarrow P' \rightarrow$ secondary minima

$$a \sin \theta = (2n+1) \frac{\lambda}{2}$$

$\rightarrow P' \rightarrow$ secondary maxima

Single slit diffraction:-



$$\Delta AOM \sim$$

$$\angle OAM = \theta, \quad \angle AOM = 90^\circ - \theta,$$

ΔAOM & ΔABN - এরা সদৃশকোণী,

সে,

$$\frac{AO}{AB} = \frac{OM}{BN}$$

$$\Rightarrow \frac{a/2}{a} = \frac{x}{\lambda}$$

$$\Rightarrow x = \frac{\lambda}{2}$$

$$AB = a = BN$$

$$\Rightarrow AO = a/2$$

$$\& OM = x$$

$$\text{Now, } \sin \theta = \frac{BN}{AB}$$

$$\Rightarrow BN = AB \sin \theta$$

$$\Rightarrow \Delta x = a \sin \theta$$

$$\Rightarrow BN = \lambda \left[\text{minima} \right]_{n=1}$$

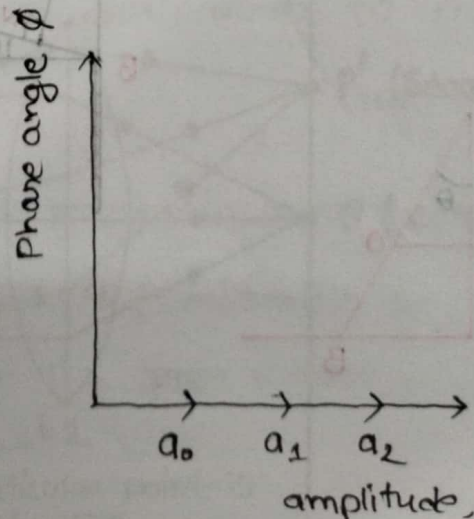
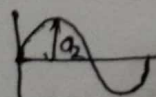
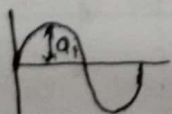
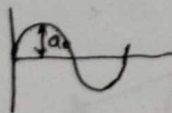
এখন, p' maxima হবে নাকি minima হবে তা নির্ণয় করে intensity-এর উৎস।

★ intensity max হলে $\rightarrow$ maxima,

★ " min " $\rightarrow$ minima.

Intensity মাপতে হলে resultant amplitude জানতে হবে.

vector polygon method:



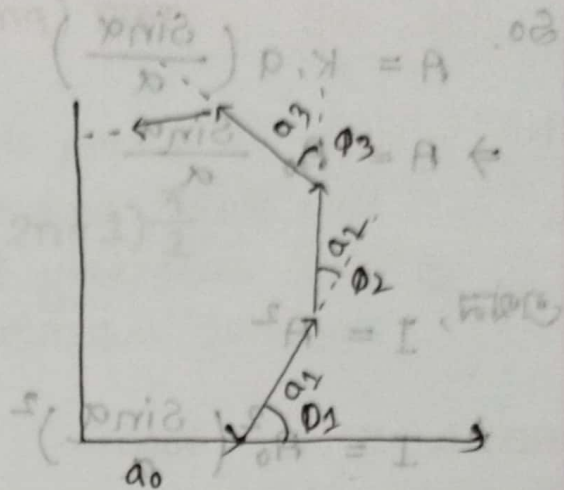
But,

① a_0

ϕ_1
② a_1

ϕ_2
③ a_2

ϕ_3
④ a_3



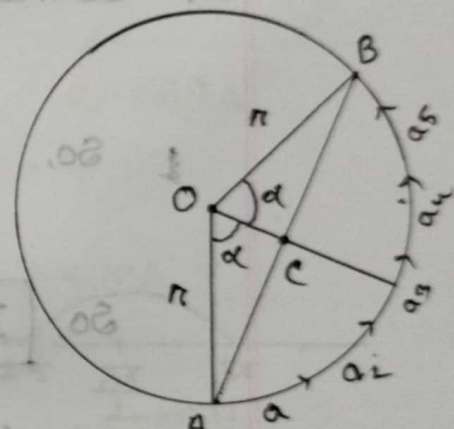
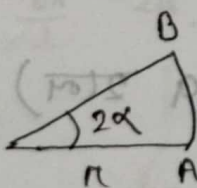
if finally it can be considered as a circle.

From BN $\Rightarrow$

$$2\alpha = \frac{2\pi}{\lambda} \times a \sin \theta$$

$$\therefore \alpha = \frac{\pi}{\lambda} \sin \theta$$

①



$$AC = BC$$

$$\therefore AB = 2AC$$

$$\theta \text{ in } \text{rad} \times \frac{\pi}{k} \Rightarrow A = 2AC$$

$\rightarrow$ resultant amplitude

$$\text{Arch} = \pi 2\alpha$$

$$\Rightarrow 2\alpha = \frac{\text{Arch}}{\pi}$$

$\pi \rightarrow k$ wavelength wave

$$\Rightarrow 2\alpha = \frac{kq}{\pi}$$

$$\Rightarrow 2\pi = \frac{kq}{\alpha}$$

③

$$\pi \alpha = \sin \alpha = \frac{AC}{AO}$$

$$\Rightarrow AC = AO \sin \alpha$$

$$\Rightarrow AC = \pi \sin \alpha$$

$$\therefore A = 2\pi \sin \alpha$$

④

So, $A = K \cdot a \left(\frac{\sin \alpha}{\alpha} \right)$

$\Rightarrow A = A_0 \frac{\sin \alpha}{\alpha}$

এখন, $I = A^2$

$I = A_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2$

$\Rightarrow I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$

in central maxima, in point P
phase diff $\alpha \rightarrow 0$

So, $\lim_{\alpha \rightarrow 0} \frac{\sin \alpha}{\alpha} = 1$

So, $I = I_0$

Now, P' (secondary minima (ল))

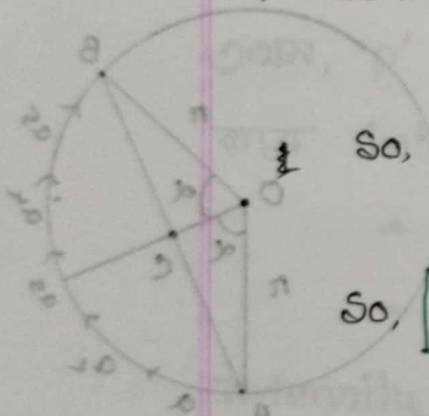
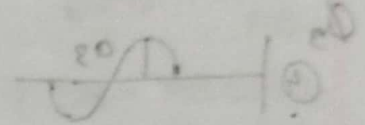
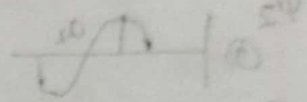
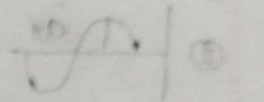
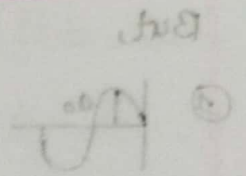
$a \sin \theta = \pm n \lambda$

$\therefore \alpha = \frac{\pi}{\lambda} \times a \sin \theta$

$\Rightarrow \alpha = \pm n \pi$

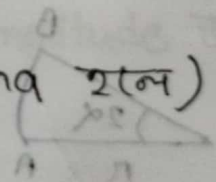
$\therefore I = I_0 \left(\frac{\sin(n\pi)}{n\pi} \right)$

$I = 0$



$\sin \theta = \frac{\lambda}{a} \Rightarrow \theta = \sin^{-1} \left(\frac{\lambda}{a} \right)$

$\theta = \sin^{-1} \left(\frac{\lambda}{a} \right)$



$\sin \theta = \frac{\lambda}{a} \Rightarrow \theta = \sin^{-1} \left(\frac{\lambda}{a} \right)$

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$\sin \theta = \frac{\lambda}{a} \Rightarrow \theta = \sin^{-1} \left(\frac{\lambda}{a} \right)$

Now, P' (Secondary maxima) हल,

$$a \sin \theta = (2n+1) \frac{\lambda}{2}$$

$$\text{So, } \alpha = \frac{\pi}{\lambda} (2n+1) \frac{\lambda}{2} = (2n+1) \frac{\pi}{2}$$

$$n=1 \text{ हल, } \alpha = \frac{3\pi}{2}$$

$$I = I_0 \left(\frac{\sin \frac{3\pi}{2}}{\frac{3\pi}{2}} \right)^2$$

$$I = 4.57 \times I_0$$

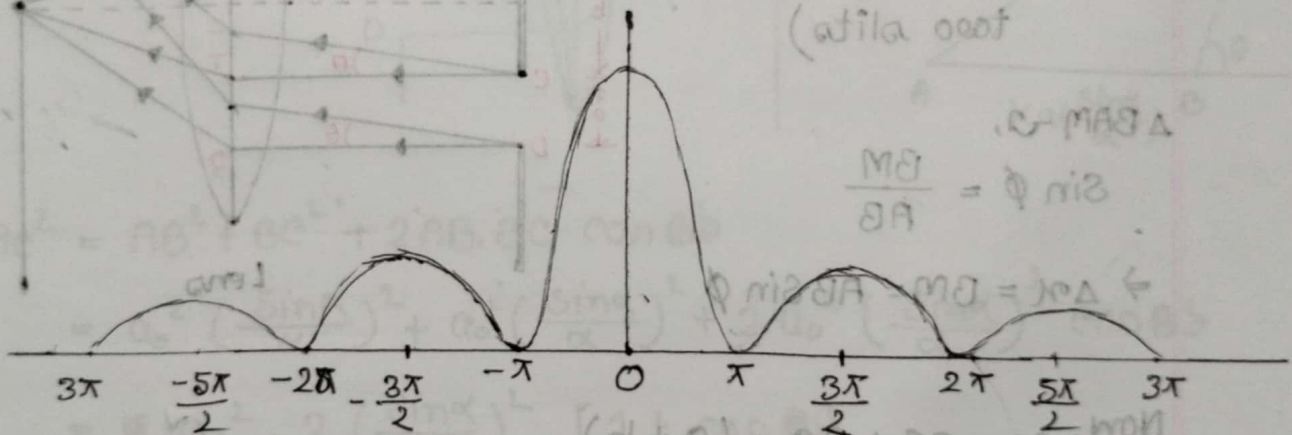


Fig: Intensity vs phase diff. graph.

$$\begin{aligned} \text{(maxima, int.) } \Delta r &= n\lambda = \theta \sin \theta (d+\Delta) \\ \text{(minima, int.) } \frac{\lambda}{2} (2n+1) &= \theta \sin \theta (d+\Delta) \end{aligned}$$

Double slit diffraction (minimax problem)

- NB:
- ① যখন secondary wave individual slit এর ক্ষর্য দিয়ে Pass করতে $\rightarrow$ diffraction হবে।
 - ② যখন secondary wave দুইটা slit এর ক্ষর্য দিয়ে যাবে তখন এবং superimpose করতে $\rightarrow$ interference হবে।

AB & CD are two rectangular slits having equal width 'a',
 $BC = b$ (spacing between two slits)

$\triangle BAM - \triangle$

$$\sin \phi = \frac{BM}{AB}$$

$$\Rightarrow \Delta x = BM = AB \sin \phi$$

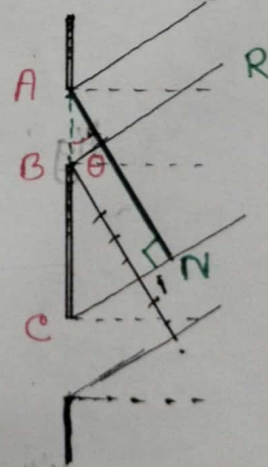
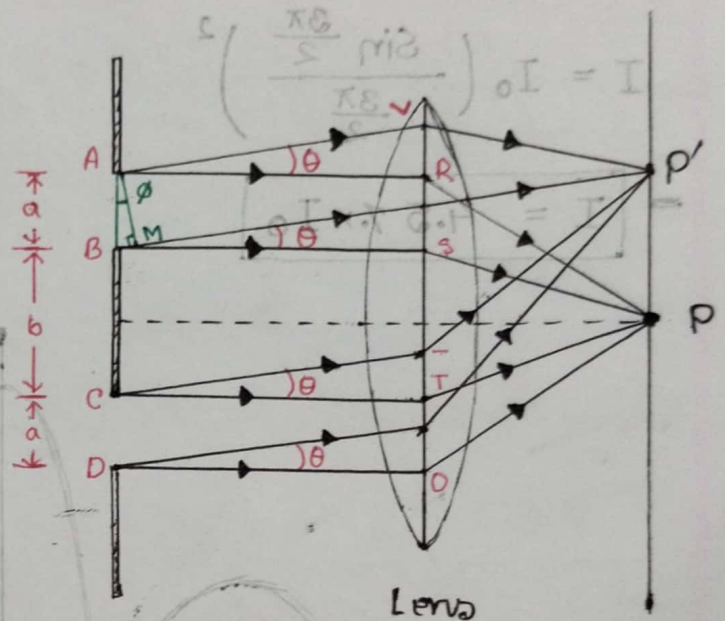
Now, $AC = AB + BC = (a + b)$

$\triangle CAN - \triangle$ $\sin \theta = \frac{CN}{AC}$

$$\Rightarrow CN = AC \sin \theta$$

$$\therefore (a+b) \sin \theta = n\lambda \text{ (intf, maxima)}$$

$$(a+b) \sin \theta = (2n+1) \frac{\lambda}{2} \text{ (intf, minima)}$$



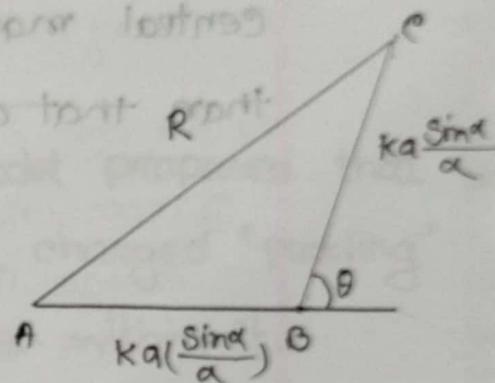
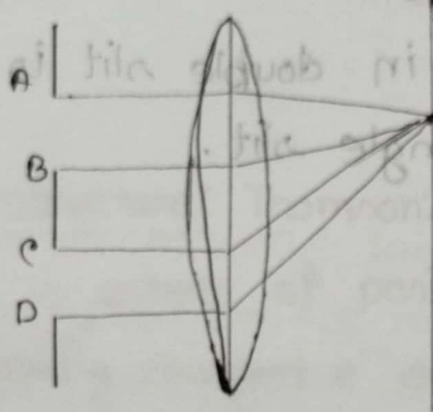
$$I = 4 I_0 \cos^2 \left(\frac{\alpha}{2} \right)$$

$$\Rightarrow I = I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2$$

$$\Rightarrow I = 4 I_0 \underbrace{\left(\frac{\sin \alpha}{\alpha} \right)^2}_{\text{diff}} \cdot \underbrace{\cos^2 \beta}_{\text{interference}}$$

if $(\theta = \phi) \rightarrow$ interference'

maxima point and diffraction minima point will be of same order.



$$AC^2 = AB^2 + BC^2 + 2 AB \cdot BC \cos \theta$$

$$= a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 + a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 + 2 a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cos \theta$$

$$= 2 a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 [1 + \cos \theta]$$

$$\Rightarrow 2 a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \left(2 \cos^2 \frac{\theta}{2} - 1 + 1 \right)$$

$$\Rightarrow 2 a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot 2 \cos^2 \frac{\theta}{2}$$

$$\Rightarrow 4 a_0^2 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \cos^2 \frac{\theta}{2}$$

$$\Rightarrow R^2 = I = 4 I_0 \left(\frac{\sin \alpha}{\alpha} \right)^2 \cdot \cos^2 \beta \quad \left[\frac{\theta}{2} = \beta \right]$$

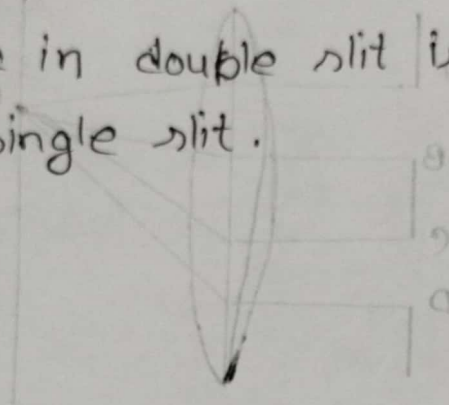
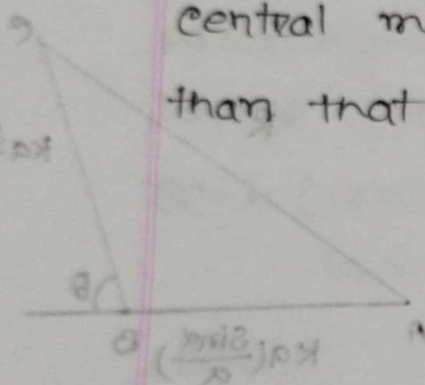
$$\alpha = \frac{\pi}{\lambda} (a \sin \theta)$$

$$\beta = \frac{\pi}{\lambda} (a+b) \sin \theta$$

At P point, $\alpha = 0$, $\beta = 0$,

$$\text{So, } I = 4 I_0$$

central maxima in double slit is 4 times higher than that of single slit.



$$I = I_1 + I_2 + I_3 + I_4 = 4 I_0$$

$$I = \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 + \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 + \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 + \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 =$$

$$[I_0 + I_0 + I_0 + I_0] \cdot \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 =$$

$$[4 + 4 - \frac{4}{2} \cos \theta] \cdot \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 =$$

$$\frac{4}{2} \cos \theta \cdot \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 =$$

$$\frac{4}{2} \cos \theta \cdot \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 =$$

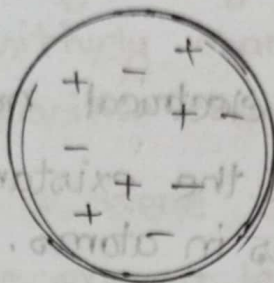
$$[I = \frac{4}{2}] \cdot \left(\frac{a \sin \theta}{\lambda} \right)^2 I_0 = I = 4 I_0$$

Atomic Model

①

Thomson atom model:

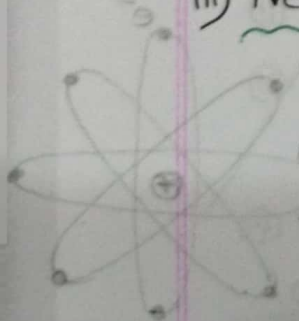
Heavy sphere of positive charge & peppered with e^- to make it electrically charge neutral.



Atomic dimension
 $= 10^{-10} \text{ m.}$

Postulates:

- i) Homogeneous structure: Thomson's model proposed that the atom is sphere of positively charged 'pudding' with negatively charged e^- embedded within it. Somewhat like raisins in a plum pudding. In other word positive charge was throughout thought to be uniformly distributed throughout the atom.
- ii) Electron arrangement: Electrons were distributed in a way that they were evenly spread out within the positive charged sphere.
- iii) Neutral atom: The model aimed to explain the neutrality of atoms since the positive and negative charges were assumed to be evenly distributed the atom as a whole was electrically neutral.



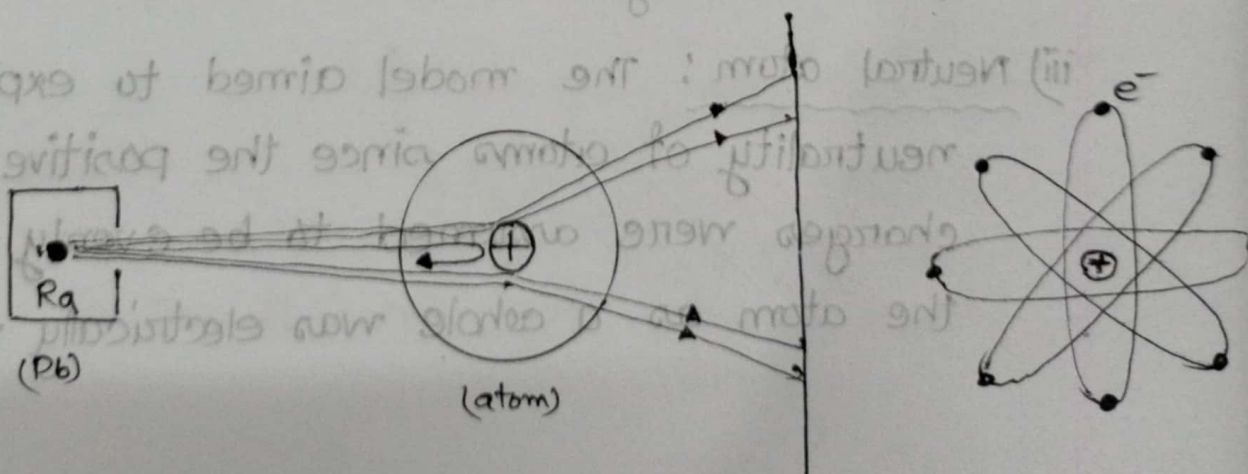
Success:

- i) First model to incorporate electrons as fundamental particle.
- ii) Explained the electrical neutrality of atoms.
- iii) Accounted for the existence of both positive and negative charges in atoms.
- iv) Provided a framework for understanding cathode-ray experiments.

Drawbacks:-

- i) Thomson's model failed to explain how the positive charge holds on the electrons inside the atom.
- ii) Failed to explain an atom's stability.
- iii) Did not mention anything about the nucleus of an atom.
- iv) Unable to explain the scattering experiment of Rutherford.

Rutherford model:



Acknowledgements:-

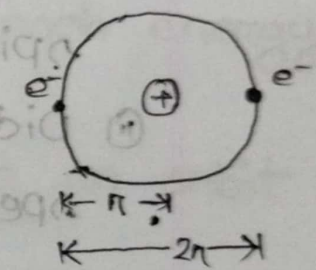
- ① The atom is mostly empty space, with a tiny dense nucleus is positively charged. Nearly all of the atom's mass & positive charge is concentrated in it.
- ② Negatively charged e^- were thought to orbit the nucleus in specific energy levels or orbit, which is compared with planets around the sun.
- ③ Neutral state due to equal number of positive and negative charge.
- ④ From a particle exp:

$$F_A = \frac{e^2}{2\pi\epsilon_0 r^2}$$

$$F_R = \frac{mv^2}{r}$$

$$F_A > F_R$$

∴ It is not stable.



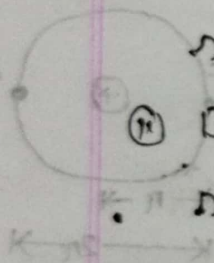
④

Success:

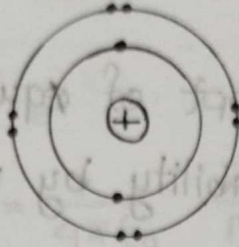
- ① Successfully explained the large-angle scattering of alpha particles
- ② Introduced the concept of atomic nucleus.
- ③ correctly predict that most of alpha particles could pass through atoms undeflected
- ④ Established the nuclear model of the atoms.

Drawbacks:

- ① According to the classical electromagnetic theory, orbiting e^- should ~~be~~ emit radiation & eventually spiral into the nucleus.
- ② Did not mention about a explanation for why e^- occupy specific energy levels and why only certain orbits were allowed.
- ③ Did not explain about spectrum.

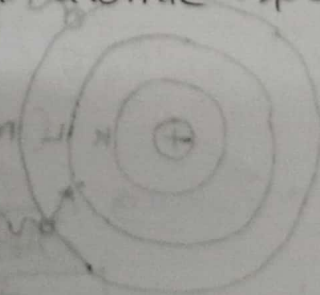
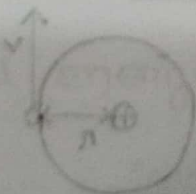


Bohr's Model :-



Acknowledgement :-

- i) e^- orbit the nucleus in well-defined energy levels which is represented by quantum numbers $(n = 1, 2, 3, 4, \dots)$
- ii) Electrons move in circular orbits within each energy level.
- iii) The angular momentum of an electron in an orbit is quantized, $mvr = \frac{nh}{2\pi}$.
- iv) e^- have lower energy closer nucleus & higher energy farther from the nucleus.
- v) e^- can change orbits by emitting & absorbing photons. This results in the observation of discrete spectral lines in atomic spectra.



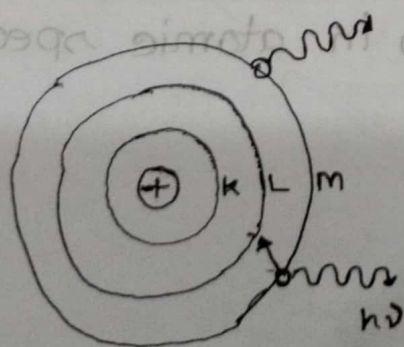
$$E_K < E_L < E_M$$

Success:-

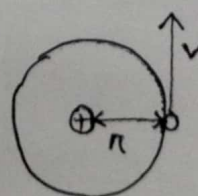
- ① Successfully explained the hydrogen spectrum & Balmer series.
- ② Introduced the concept of quantized energy levels.
- ③ Explained atomic stability by postulating non-radiating orbits.
- ④ Provided accurate predictions for hydrogen ionization energy.
- ⑤ Laid the foundation for quantum mechanics.

Limitations:-

- ① Limited hydrogen like species (atoms with a single electron) & did not fully explain the behaviour of multi-electron atoms.
- ② did not provide a detailed description of the e^- motion as a wave, which is a key aspect of quantum mechanics.
- ③ can not explain Zeeman & Stark effect.



$$E_K < E_L < E_M$$



As e^- orbits around nucleus, so centrifugal force is arises and it provides static attraction coulomb force.

$$F_c = F_e$$

$$\text{or } \frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{ze^2}{r^2}$$

$$\Rightarrow v^2 = \frac{ze^2}{4\pi\epsilon_0 m r} \quad (1)$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2 \text{ m}^{-1} \text{ N}^{-1}$$

$$\Rightarrow \frac{n^2 h^2}{4\pi^2 m^2 r^2} = \frac{ze^2}{4\pi\epsilon_0 m r}$$

$$\Rightarrow r_n = \frac{n^2 h^2 \epsilon_0}{\pi m z e^2}$$

$$r_1 = 0.531 \text{ \AA}$$

$$\Rightarrow r_n = 0.531 \times \frac{n^2}{z}$$

Kinetic energy, $E_k = \frac{1}{2} m v^2$

$$= \frac{1}{2} m \frac{ze^2}{4\pi\epsilon_0 m r}$$

$$\therefore E_k = \frac{ze^2}{8\pi\epsilon_0 r}$$

Potential energy, $E_p = v \times (-e)$

$$= -\frac{ze^2}{4\pi\epsilon_0 r}$$

$$E_p = -\frac{ze^2}{4\pi\epsilon_0 r}$$

8

Total energy,

$$E = E_k + E_p$$

$$= \frac{ze^2}{8\pi\epsilon_0 r} - \frac{ze^2}{4\pi\epsilon_0 r}$$

$$= -\frac{ze^2}{8\pi\epsilon_0 r}$$

$$= -\frac{ze^2}{8\pi\epsilon_0} \times \frac{\pi m z e^2}{n^2 h^2 \epsilon_0}$$

$$E_n = -\frac{m e^4 z^2}{8 \epsilon_0^2 n^2 h^2}$$

$$E_1 = -13.6 \text{ eV}$$

$$n=1$$

$$z=1$$

$$E_n = -13.6 \times \frac{z^2}{n^2}$$

$$\Delta E = E_2 - E_1$$

$$\Rightarrow \frac{hc}{\lambda} = E_2 - E_1$$

So, Now,

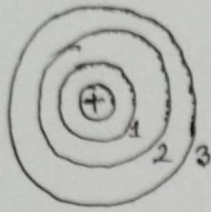
$$-\frac{m z^2 e^4}{8 n_2^2 h^2 \epsilon_0^2} + \frac{m z^2 e^4}{8 n_1^2 h^2 \epsilon_0^2} = \frac{hc}{\lambda}$$

$$\text{or, } \frac{1}{\lambda} = \frac{m z^2 e^4}{8 h^3 \epsilon_0^2 c} \left(\frac{1}{n_2^2} - \frac{1}{n_1^2} \right)$$

$$\text{Rydberg constant : } R_H = \frac{9.11 \times 10^{-31} \times 1 \times (1.6 \times 10^{-19})^4}{8 \times (6.626 \times 10^{-34})^3 \times (8.854 \times 10^{-12})^2 \times 3 \times 10^8}$$

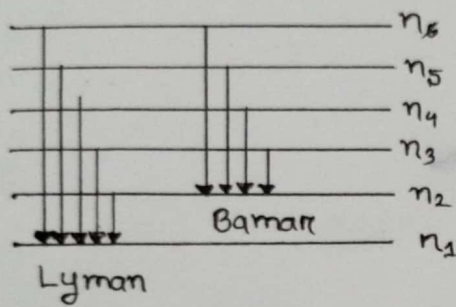
$$= 1.091 \times 10^8 \text{ m}^{-1}$$

Electron Transition series:-



$n_1 = 1$
 $n_2 = 2, 3, 4, \dots$ → Lyman Series

$n_1 = 2$
 $n_2 = 3, 4, 5, \dots$ → Balmer Series



$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$$

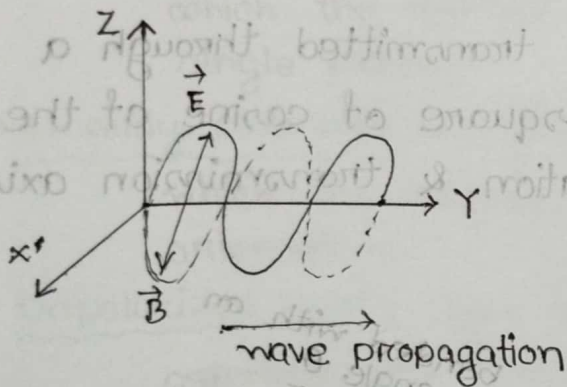
$n = 1 \rightarrow$ Lyman
 $n = 2 \rightarrow$ Balmer
 $n = 3 \rightarrow$ Paschen
 $n = 4 \rightarrow$ Brackett
 $n = 5 \rightarrow$ Pfund
 $n = 6 \rightarrow$ Humphreys

Polarization

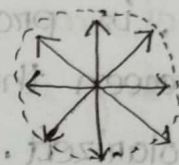
(Pol-1)

Polarization :-

Polarization is a property of transverse wave which specifies the geometrical orientation of the oscillation. In a transverse wave, the direction of the oscillation is perpendicular to the direction of motion of the wave.



unpolarized light:



Symmetrical Electric field divided in two planes (axes)

Let consider,



⇒ indicates parallel



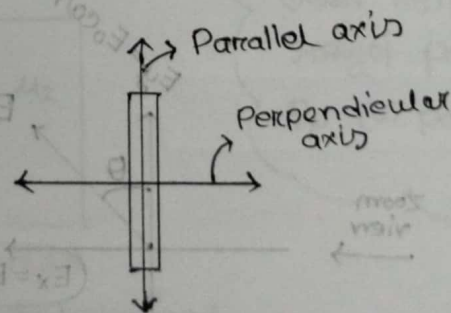
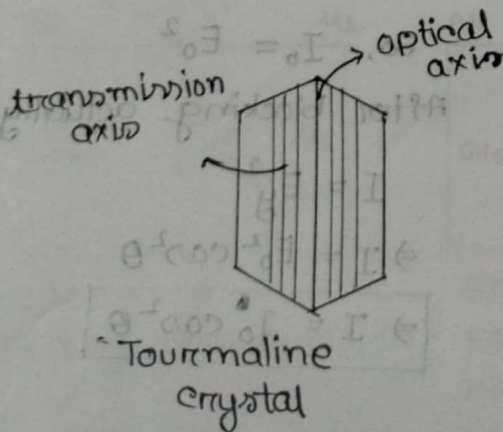
⇒ indicates perpendicular

so, unpolarized Light is divided into two components:

- ① Parallel.
- ② perpendicular.

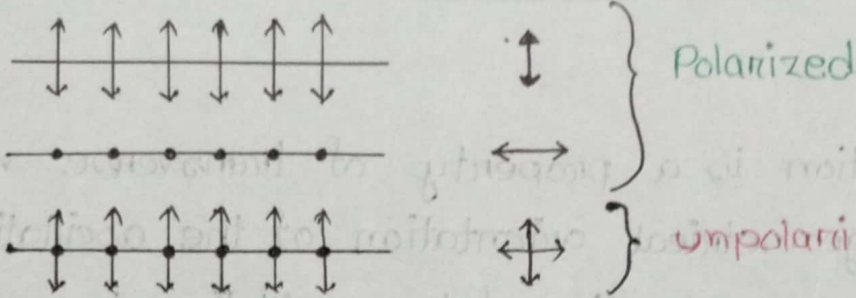
- * if we block parallel → perpendicular remains
- * if we block perpendicular → Parallel remains

Polarized Light



Here, perpendicular is blocked, Parallel passes / vibrates.

Polarized Light.

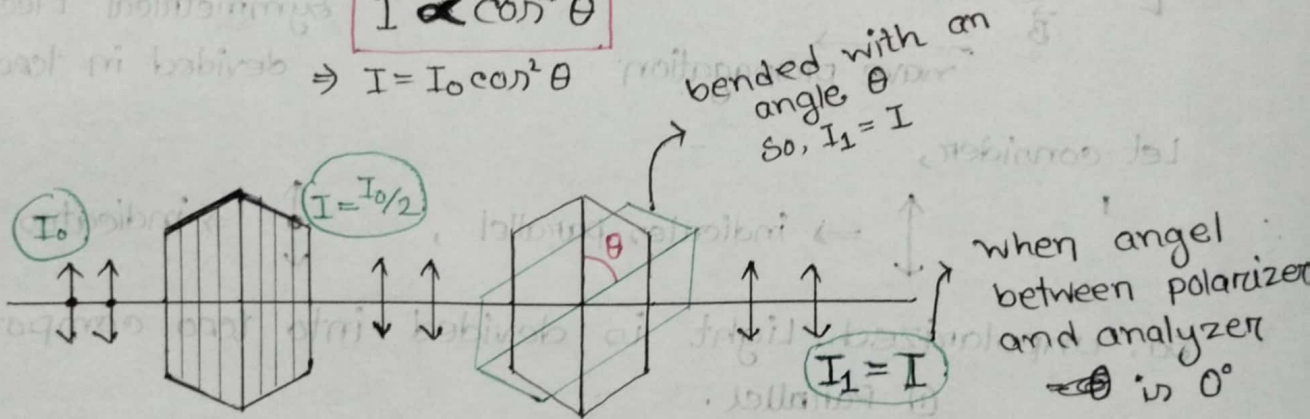


Malus Law / cosine law:

The Intensity of polarized light transmitted through a polarizer, is proportional to the square of cosine of the angle between the plane of polarization & transmission axis of the polarizer.

$$I \propto \cos^2 \theta$$

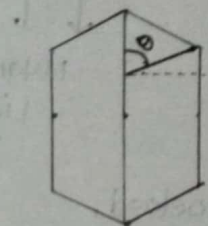
$$\Rightarrow I = I_0 \cos^2 \theta$$



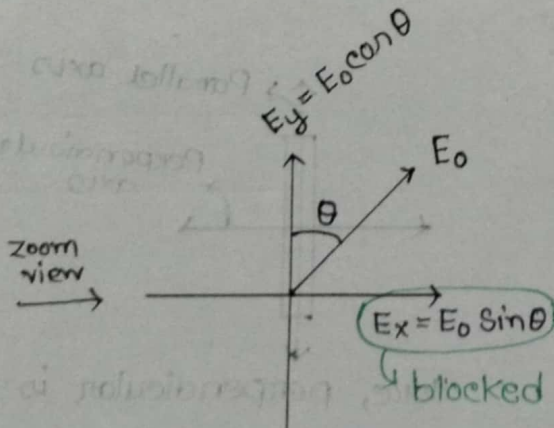
intensity of unpolarized Light = I_0 ,

intensity of polarized light = $I = I_0/2$

$$I_1 = \frac{I_0}{2} \cos^2 \theta$$



Tourmaline crystal



$$\text{So, } I_0 = E_0^2$$

After blocking actually,

$$I = E_y^2$$

$$\Rightarrow I = E_0^2 \cos^2 \theta$$

$$\Rightarrow I = I_0 \cos^2 \theta$$

৩ টা way-তে unpolarized light-কে Polarized করা হয় :

(i) by reflection

(ii) by refraction (using pile of glass plate)

(iii) double refraction. (Calcite crystal)

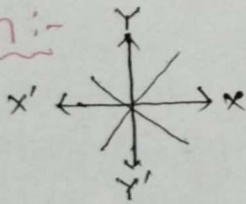
Polarization due to reflection

* Plane polarised: A plane polarized light where wave is wave in which the electric vector is everywhere confined to a single plane.

* Linearly polarised: A Linearly polarised light wave is wave in which the electric vector oscillates in a given constant orientation.

* Unpolarized light: Light in which the plane of vibration are symmetrically distributed about the propagation direction of the wave is known as the unpolarized light.

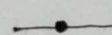
reflection:-



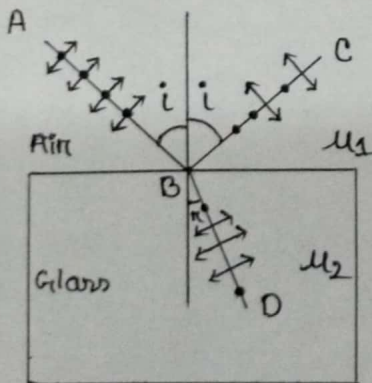
Now,



Parallel to YY' axis



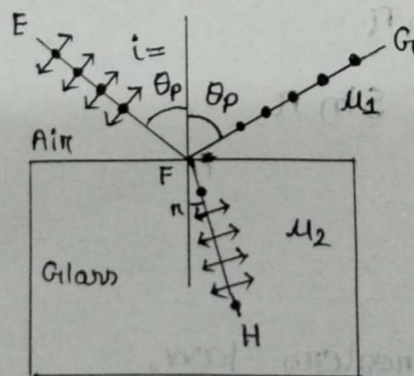
Perpendicular to YY' axis.



AB → unpolarized

BC → partially polarized

BD → " "



EF = unpolarized

FG = completely polarized

FH = partially polarized

i যখন একটা Particular angle (θ_p) এর সমান হয় তখন reflected light সম্পূর্ণ polarized হয়। θ_p -কে polarizing angle বোলে।

Snel's Law,

$$\frac{\sin \theta_p}{\sin \pi} = \frac{\mu_2}{\mu_1}$$

- * As the incident angle is varied the extend of polarisation of reflected ray varies.
- * at particular angle θ_p , Parallel (P) component goes 0,
- * It contains only perpendicular (S) component.
- * θ_p is the polarizing angle.

But -

- * intensity is 15% of incident S-component.
- * 85% is refracted having (S+P) component.

Brewster's Law :-

"The tangent of the angle at which polarization is obtained by reflection is numerically equal to the refractive index of the medium."

So, $\tan \theta_p = \frac{\mu_2}{\mu_1}$

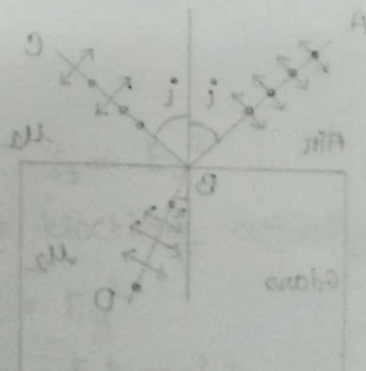
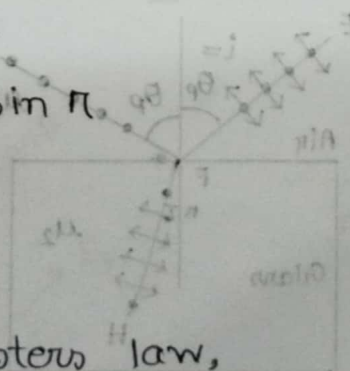
$$\Rightarrow \frac{\sin \theta_p}{\cos \theta_p} = \frac{\sin \theta_p}{\sin \pi}$$

$$\Rightarrow \cos \theta_p = \sin \pi$$

$$\Rightarrow \sin (90 - \theta_p) = \sin \pi$$

$$\Rightarrow 90 - \theta_p = \pi$$

$$\Rightarrow \pi + \theta_p = 90^\circ$$



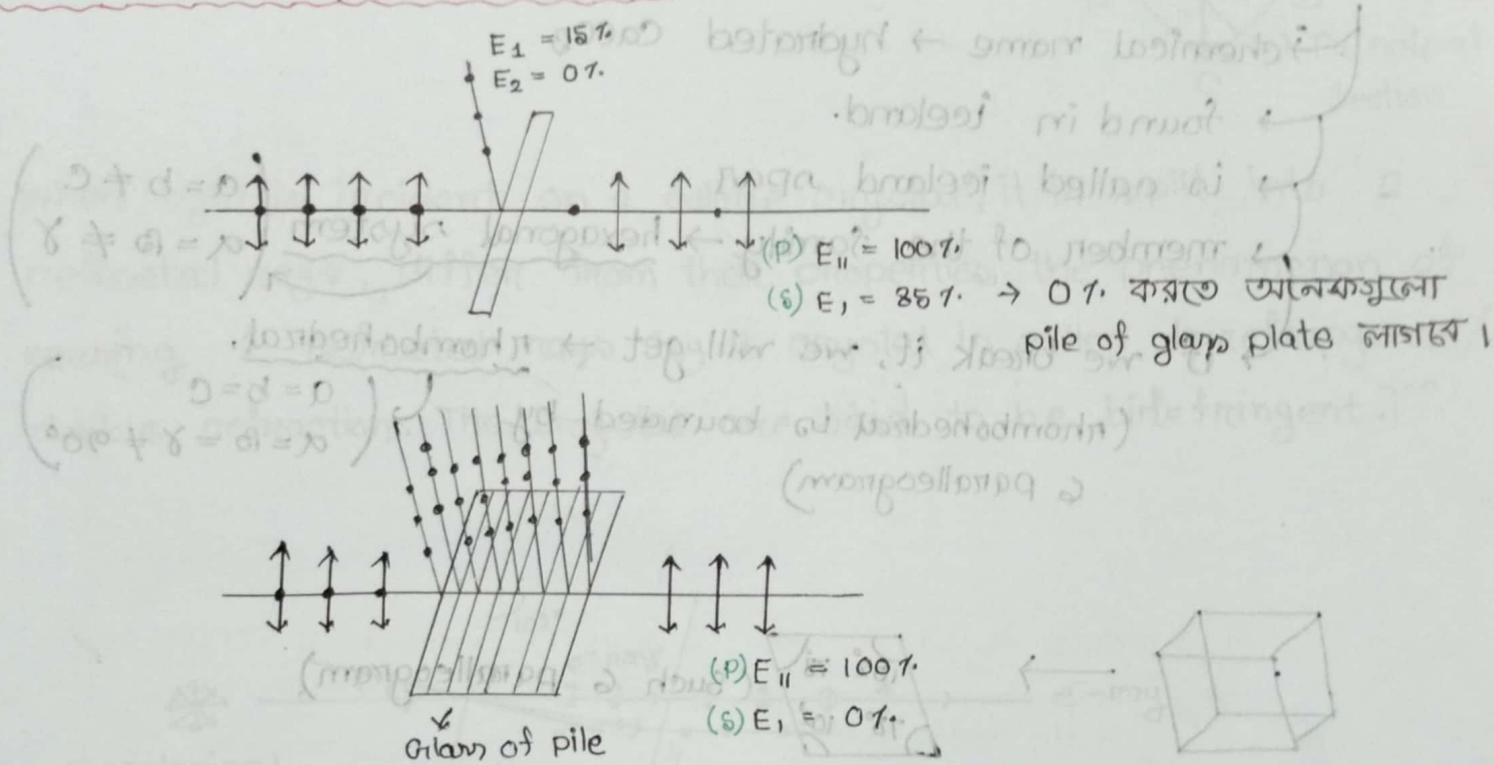
Q1 State and extends Brewster's law,

or,

show that reflected and refracted rays are 90° apart.

✓ polarization by refraction:

(pol-2)



- * If a stack of glass plates is used instead of a single plate, refraction from successive surfaces occurs leading to the filtering of the s-component in the transmitted ray.
- * Ultimately the transmitted ray consists of p-component alone.

$I_p \rightarrow$ p-component intensity

$I_s \rightarrow$ s-component "

The degree of polarization,

$$p = \frac{I_p - I_s}{I_p + I_s} = \frac{m}{m + \left(\frac{2\mu}{1 - \mu^2} \right)^2}$$

where,

$m =$ the number of plates

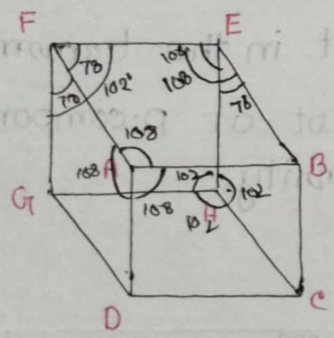
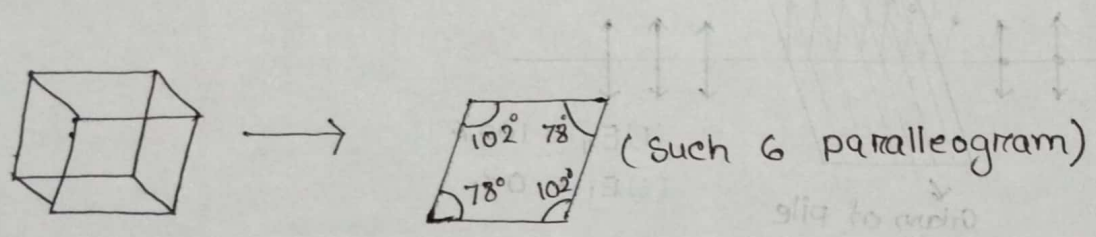
$\mu =$ refractive index of the material

But,

The method of polarizing light is also not efficient because a good portion of light is lost in reflections.

calcite crystal

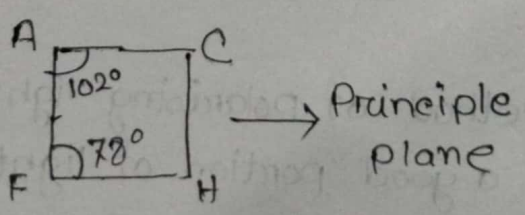
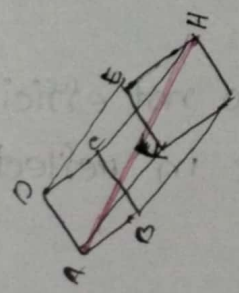
- chemical name → hydrated CaCO3
- found in iceland.
- is called iceland spar
- member of the family → hexagonal system $\begin{pmatrix} a=b \neq c \\ \alpha=\beta \neq \gamma \end{pmatrix}$
- if we break it, we will get → rhombohedral,
(rhombohedral is bounded by 6 parallelogram) $\begin{pmatrix} a=b=c \\ \alpha=\beta=\gamma \neq 90^\circ \end{pmatrix}$



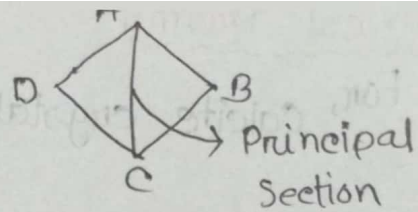
A, H → Blunt axis
 এই দুই corner-এ ও দুই per angle-র obtuse angle (102°)
 A, H → obtuse angle.

যে Plane-এ একটি optical axis আছে → uni-axial
 Ex: calcite crystal,

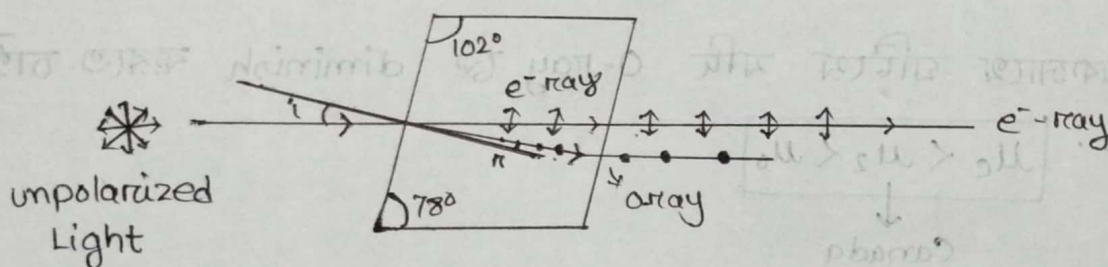
যে Plane-এ দুইটি optical axis আছে → bi-axial.



Double refraction করে $\rightarrow$ principal plane.



[When light is incident on a calcite crystal, it is split into 2 refracted rays, differ from their properties, the phenomenon of causing 2 refracted rays by a crystal is called birefringence/ double refraction. The crystal are said to be birefringent.]



* One of the ray obey's snell's law, so it is called ordinary ray (O-ray),

$$\text{So, } \angle i < \angle r \Rightarrow \frac{\sin i}{\sin r} = \frac{v_{\text{light}}}{v_{\text{heavy}}} = \mu$$

* The other ray does not obey snell's law, so it is called extra-ordinary ray (e-ray)

$$\mu_o = \frac{\sin i}{\sin r_o} = \frac{c}{v_o}$$

$$\mu_e = \frac{\sin i}{\sin r_e} = \frac{c}{v_e}$$

$$\mu_o > \mu_e, \quad r_o < r_e, \quad v_o < v_e$$

* Polarizer: A polarizer is an optical device that transforms unpolarized Light into polarized Light.

* Analyzer: An analyzer is a device, which is used to identify the direction of vibration of linearly polarised light.

Nicol Prism

For, calcite crystal,

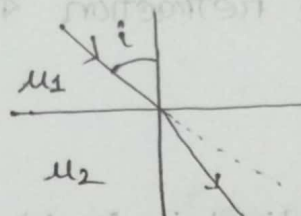
$$\mu_o = 1.658$$

$$\mu_e = 1.468$$

incident angle, critical angle এর চেয়ে

বড় হলে light পুরোটাই reflected

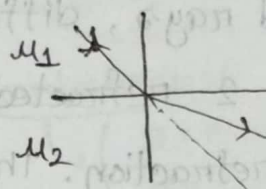
হয়ে যায়।



Case-1:-

when,

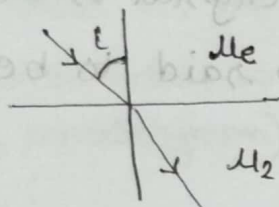
$$\mu_2 > \mu_1$$



Case-2:-

when,

$$\mu_1 > \mu_2$$



Case-2:-

when,

$$\mu_2 > \mu_e$$

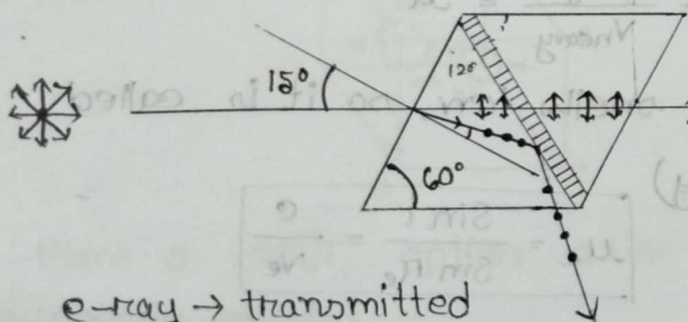
Case: 1 & 2 একসাথে ঘটিয়ে যদি O-ray কে diminish করতে চাই,

তার condition:

$$\mu_e < \mu_2 < \mu_o$$

Canada
Balsam

$$\mu_{CB} = 1.550$$



Linearly polarized Light

e-ray → transmitted

O-ray → internally reflected

$$\text{Here, } \sin \theta_c = \frac{\mu_{CB}}{\mu_o}$$

$$\Rightarrow \sin \theta_c = \frac{1.550}{1.658}$$

$$\Rightarrow \theta_c = 69.2$$

critical angle.

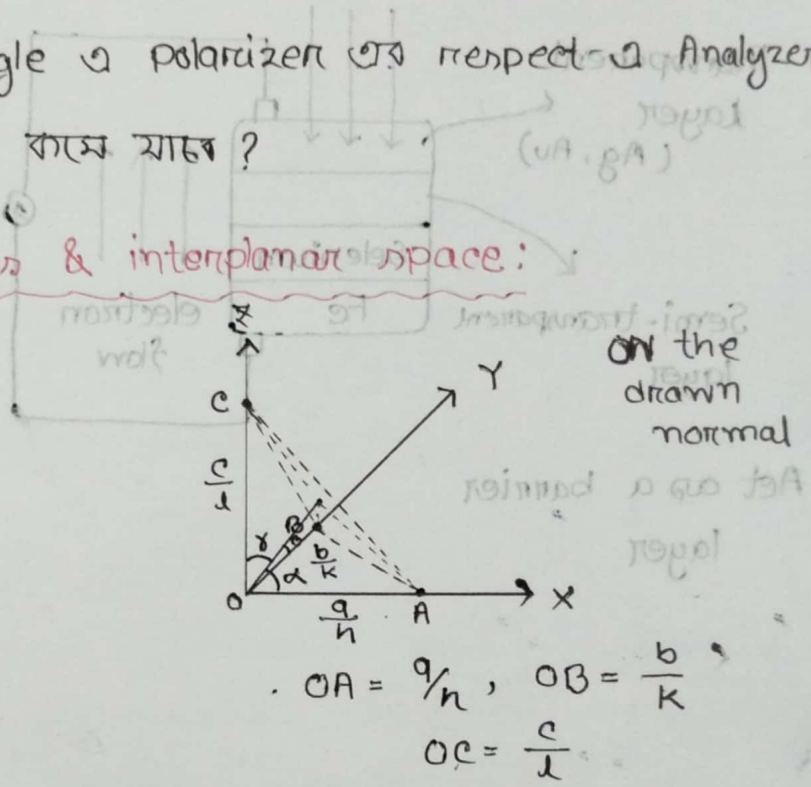
Q-1] Calcite crystal এ double refraction phenomena ব্যাখ্যা কর, and explain e-ray, o-ray.

Q-2] Nicol prism এর action explain করো;
 or, Prism কীভাবে Polarizer হিসেবে কাজ করে ?

Q-3] [Malus law] কত degree angle এ polarizer এর respect এ Analyzer কে ঘুরালে intensity 7. কমে যাবে ?

Relation between Miller indices & interplanar space:

h, k, l are miller indices
 (h, k, l) crystal plane
 intercept points: $\frac{1}{h}, \frac{1}{k}, \frac{1}{l}$
 $\downarrow \quad \downarrow \quad \downarrow$
 $x \quad y \quad z$



$$\cos^2 \alpha = \frac{d^2}{a^2/h^2}$$

$$\cos^2 \beta = \frac{d^2}{b^2/k^2}$$

$$\cos^2 \gamma = \frac{d^2}{c^2/l^2}$$

Now,
 $\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$

$$\Rightarrow d^2 \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right) = 1$$

$$\Rightarrow d = \left(\frac{h^2}{a^2} + \frac{k^2}{b^2} + \frac{l^2}{c^2} \right)^{-\frac{1}{2}}$$

Orthorhombic ($a \neq b \neq c$)

for cubic,

$$d = \left(\frac{h^2 + k^2 + l^2}{a^2} \right)^{-\frac{1}{2}}$$

$$d = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

ON, OA এর দ্বিধিকার কোণ = α
 ON, OB " " " = β
 ON, OC " " " = γ

$\Delta OAN \Rightarrow$

$$\cos \alpha = \frac{ON}{OA} = \frac{d}{a/h}$$

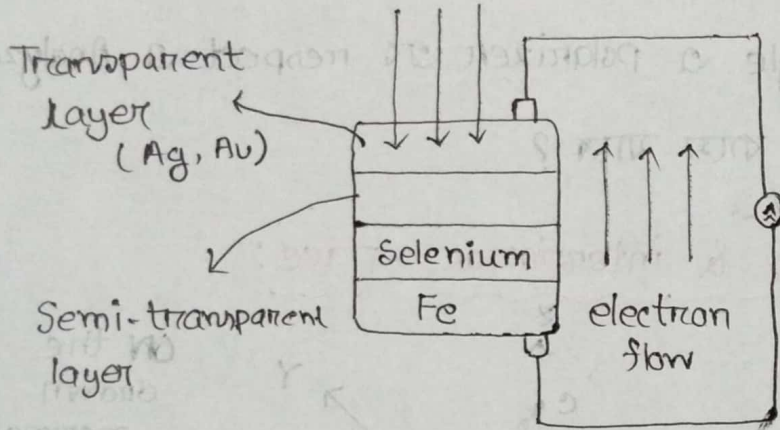
$$\cos \beta = \frac{ON}{OB} = \frac{d}{b/k}$$

$$\cos \gamma = \frac{ON}{OC} = \frac{d}{c/l}$$

$[ON = d]$
 $\downarrow$
 interplanar spacing

Photo-voltaic effect :-

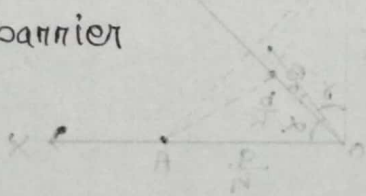
photo-voltaic cell $\rightarrow$ self generating cell.



Electron field কাজ করবে
নিচের দিকে।

e-flow হবে তাই
উপরের দিকে

Act as a barrier
layer



$$\frac{b}{N^p} = 80 \quad ; \quad \frac{b}{N^p} = 100$$

$$\frac{b}{N^p} = 80$$

$\lambda = 1000$ nm
 $\lambda = 1000$ nm
 $\lambda = 1000$ nm

$\triangle ABC$

$$\frac{b}{N^p} = \frac{100}{100} = 1000$$

$$\frac{b}{N^p} = \frac{100}{100} = 1000$$

$$\frac{b}{N^p} = \frac{100}{100} = 1000$$

$$[b = 100]$$

normalization
process

$$\frac{1}{1} + \frac{1}{1} + \frac{1}{1} = 3$$

$$\frac{b}{N^p} = 1000$$

$$\frac{b}{N^p} = 1000$$

$$\frac{b}{N^p} = 1000$$

$$L = 1000 + 1000 + 1000$$

$$L = \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10} \right) b$$

$$\frac{1}{L} \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10} \right) = b$$

(1000 + 1000 + 1000)

$$\frac{1}{L} \left(\frac{1}{10} + \frac{1}{10} + \frac{1}{10} \right) = b$$

$$\frac{1}{1000 + 1000 + 1000} = b$$